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[Intervention Review]

Blood pressure management in reperfused ischemic stroke

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ABSTRACT

Rationale

Stroke is the second leading cause of death worldwide and a major contributor to disability, with ischemic strokes accounting for 80% to 85% of all strokes. Appropriate blood pressure management during and after procedures like thrombectomy and thrombolysis is essential, as sustained high blood pressure can worsen outcomes. Guidelines recommend maintaining blood pressure below 180/105 mmHg during the 24 hours after a successful procedure; however, this recommendation is based on low-certainty evidence. Some studies suggest that lowering systolic blood pressure (SBP) to below 160 mmHg could improve outcomes and reduce the risk of intracranial hemorrhage. Given the prevalence of ischemic strokes, establishing optimal blood pressure targets is crucial for minimizing complications, enhancing recovery, and reducing the overall burden on healthcare systems.

Objectives

To assess the benefits and harms of intensive systolic blood pressure management (target less than 160 mmHg) versus conventional management (target less than 180 mmHg) in people undergoing ischemic stroke reperfusion via systemic thrombolysis or endovascular thrombectomy.

Search methods

We searched CENTRAL, MEDLINE, Embase, and two trial registers, together with reference checking, citation searching, and contact with study authors, to identify studies. The latest search date was 20 March 2025.

Eligibility criteria

We included randomized controlled trials (RCTs) in adults with acute ischemic stroke reperfused by systemic thrombolysis or endovascular thrombectomy, comparing intensive SBP management (targeting any SBP objective less than 160 mmHg) with conventional SBP management (targeting SBP less than 180 mmHg).

Outcomes

Our critical outcomes were clinical function (assessed dichotomously using the modified Rankin scale (mRS)), quality of life, and neurologic adverse events (assessed as any intracranial hemorrhage (ICH) and as symptomatic ICH). Important outcomes included clinical function (assessed continuously), neurologic status (assessed as a continuous and dichotomous outcome using the National Institutes of Health Stroke Scale (NIHSS)), all-cause mortality, hospital length of stay, and other adverse events.

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Risk of bias

We used the Cochrane RoB 2 tool to assess risk of bias.

Synthesis methods

We synthesized results for each outcome using meta-analysis, where possible, with a random-effects model. We used GRADE to assess the certainty of evidence for each outcome.

Included studies

We included nine studies with 4381 participants, primarily from upper-middle and high-income countries. Seven studies focused on participants who underwent endovascular thrombectomy, while two studies focused on participants who received systemic thrombolysis. Five studies aimed for a target blood pressure of < 140 mmHg, two aimed for < 120 mmHg, one for < 160 mmHg, and another study included two intervention arms with different targets (< 140 mmHg and < 160 mmHg). Eight studies targeted blood pressure < 180 mmHg and one study targeted blood pressure < 160 mmHg in the conventional blood pressure management group.

Synthesis of results

Intensive blood pressure management versus conventional blood pressure management

Critical outcomes

Intensive SBP management results in little to no clinically meaningful difference in clinical function (dichotomous outcome) (risk ratio (RR) 0.89, 95% confidence interval (CI) 0.80 to 0.98; $I^2 = 51\%$; 9 studies, 4272 participants; high-certainty evidence). Intensive SBP management likely results in little to no clinically meaningful difference in quality of life (standardized mean difference (SMD) -0.14, 95% CI -0.49 to 0.22; $I^2 = 75\%$; 3 studies, 3165 participants; moderate-certainty evidence). For any ICH, intensive SBP management does not increase or reduce neurologic adverse events (RR 0.99, 95% CI 0.87 to 1.14; $I^2 = 30\%$; 9 studies, 4304 participants; high-certainty evidence). For symptomatic neurologic adverse events (ICH), intensive SBP management may result in little to no clinically meaningful difference (RR 1.03, 95% CI 0.77 to 1.36; $I^2 = 2\%$; 9 studies, 4304 participants; low-certainty evidence).

Important outcomes

Intensive SBP management likely results in little to no clinically meaningful difference in clinical function (continuous outcome) (mean difference (MD) 0.28, 95% CI -0.20 to 0.77; $I^2 = 30\%$; 4 studies, 787 participants; moderate-certainty evidence). It also probably results in little to no clinically meaningful difference in neurologic status (continuous) (MD 1.77, 95% CI 0.37 to 3.16; $I^2 = 10\%$; 6 studies, 1710 participants; moderate-certainty evidence) and it probably reduces the number of participants with a favorable neurologic status (dichotomous) (RR 0.71, 95% CI 0.51 to 1.01; $I^2 = 0\%$; 3 studies, 713 participants; moderate-certainty evidence). Additionally, intensive SBP management probably increases all-cause mortality (RR 1.19, 95% CI 1.08 to 1.32; $I^2 = 0\%$; 9 studies, 4297 participants; moderate-certainty evidence). It also probably results in little to no clinically meaningful difference in hospital length of stay (MD 0.38, 95% CI -1.40 to 2.15; $I^2 = 33\%$; 5 studies, 3086 participants; moderate-certainty evidence) and it may have little to no clinically meaningful effect on other adverse events (RR 1.21, 95% CI 0.86 to 1.72; $I^2 = 61\%$; 8 studies, 4012 participants; very low-certainty evidence).

Overall, the certainty of the evidence ranged from high to very low across outcomes, with downgrading mainly due to risk of bias, imprecision, and inconsistency; certainty was lowest for symptomatic neurologic adverse events and other adverse events.

Authors' conclusions

Intensive systolic blood pressure management likely results in little to no clinically meaningful differences in quality of life, clinical function (continuous outcome), neurologic status (continuous outcome; NIHSS), or hospital length of stay. It also results in little to no clinically meaningful difference in clinical function (dichotomous) and does not increase or reduce neurologic adverse events. Additionally, it may result in little to no clinically meaningful difference in symptomatic neurologic adverse events and other adverse events. Lastly, it probably reduces the number of participants with a favorable neurologic status (dichotomous outcome) and probably increases all-cause mortality. The certainty of the evidence ranged from very low to high.

New studies must identify which patients benefit from different blood pressure strategies through subgroup analyses based on age, baseline blood pressure, and stroke severity, while reporting sex-disaggregated data. Including imaging and physiological markers can help set individualized blood pressure targets. Future trials should involve participants from low- and middle-income countries and consider factors such as access to reperfusion therapy and drug availability.

Funding

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Registration

Protocol (2025) available via doi.org/10.1002/14651858.CD016085.

PLAIN LANGUAGE SUMMARY

What are the benefits and risks of lowering blood pressure more intensively compared with standard blood pressure control after treatment to restore blood flow in people with a stroke?

Key messages

- In adults who had treatment to reopen a blocked brain artery, lowering blood pressure very intensively (usually below 160 mmHg) does not improve the chances of living independently three months after the stroke compared with standard blood pressure targets (usually below about 180 mmHg). “Living independently” means being able to walk and manage most everyday activities without help.
- Intensive blood pressure lowering probably increases the risk of death and poor recovery after stroke. Differences in brain bleeding between intensive and standard blood pressure strategies were small.
- Most of the people in the studies were treated in hospitals in upper-middle and high-income countries with specialized stroke services. We do not know whether the results would be the same in countries with fewer resources or in all groups of people who have an ischemic (blood blockage) stroke.

What is an ischemic stroke?

A stroke occurs when the blood supply to part of the brain is suddenly interrupted. In an ischemic stroke, this interruption is caused by a blood clot blocking an artery in the brain. It can lead to various neurological problems, such as weakness, decreased sensitivity, lack of coordination, and difficulty speaking.

How is a stroke treated?

Doctors can sometimes reopen the blocked artery using clot-busting medicines given into a vein, or by performing a procedure to remove the clot with a thin tube passing through the blood vessels (often called mechanical thrombectomy). These are called reperfusion treatments because they restore blood flow to the brain.

After reperfusion, many people have very high blood pressure, which can increase the risk of bleeding in the brain. However, lowering blood pressure too much or too quickly can reduce blood flow to brain tissue that is still at risk and be harmful to the brain.

What did we want to find out?

We wanted to find out whether intensive blood pressure lowering after reperfusion treatment, compared with standard blood pressure control:

- improves recovery and ability to live independently;
- improves health-related quality of life;
- improves survival;
- increases or reduces bleeding in the brain;
- decreases the duration of days in hospital;
- affects the frequency of other unwanted effects.

What did we do?

We searched for clinical studies (in which people are assigned to groups by chance) that included adults with ischemic stroke who had received reperfusion treatment with a clot-busting medicine, mechanical thrombectomy, or both, and compared more intensive blood pressure targets (less than 160 mmHg) with standard targets (less than 180 mmHg) shortly after reperfusion.

We combined and summarized the results of the studies. We also considered how the studies were conducted, the number of participants included, and the similarities and differences between the studies to judge how much confidence we can have in the results.

What did we find?

We found nine studies, involving 4381 adults with ischemic stroke who had reperfusion treatment. Most participants were treated in hospitals in high-income countries, with a smaller number in hospitals in middle-income countries. None of the studies took place in low-income countries. In most studies, the intensive strategy aimed to keep systolic blood pressure (the top number in a blood pressure reading) below 140 mmHg, while the standard strategy aimed to keep it below about 180 mmHg.

Main results

- Intensive blood pressure lowering makes little or no difference to the chance of living independently about three months after the stroke, compared with standard blood pressure control.
- Intensive blood pressure lowering probably makes little or no difference to health-related quality of life.

- Differences in brain bleeding (including severe bleeding that worsens symptoms) between intensive and standard blood pressure strategies were small and uncertain.
- Intensive blood pressure lowering probably increases the risk of death and probably increases the chance of a poor recovery after stroke.

Overall, the results suggest that lowering blood pressure to very low levels after reperfusion is unlikely to provide additional benefit and might cause harm in some people, compared with keeping blood pressure within more standard targets.

What are the limitations of the evidence?

The evidence has several important limitations, such as the relatively small numbers of people in some studies, which reduce the reliability of the results. Additionally, most participants were treated in specialized centers in middle and high-income countries; there is no evidence from low-income countries. Some groups of people may be under-represented (for example, older adults with many other health problems), and many studies did not provide detailed results separately for women and men.

Because of these limitations, the results should be interpreted with caution. More research is needed to determine whether specific groups of people might benefit from different blood pressure targets.

How up to date is this evidence?

The evidence in this review is up to date as of March 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Intensive systolic blood pressure management compared to conventional systolic blood pressure management in people undergoing ischemic stroke reperfusion

Intensive systolic blood pressure management compared to conventional systolic blood pressure management in people undergoing ischemic stroke reperfusion

Patient or population: people undergoing ischemic stroke reperfusion

Setting: inpatients from Australia, Brazil, Canada, Chile, China, Colombia, France, Hong Kong, India, Italy, South Korea, Russian Federation, Singapore, Spain, Taiwan, Thailand, United Kingdom, United States, Vietnam

Intervention: intensive systolic blood pressure management

Comparison: conventional systolic blood pressure management

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with conventional systolic blood pressure management	Risk with intensive systolic blood pressure management				
Clinical function (dichotomous) assessed with: modified Rankin scale follow-up: 90 days	590 per 1000	525 per 1000 (472 to 578)	RR 0.89 (0.80 to 0.98)	4272 (9 RCTs)	⊕⊕⊕⊕ High ^{a,b,c}	Intensive systolic blood pressure management results in little to no clinically meaningful difference in clinical function (dichotomous). Since no established minimum clinically important difference (MCID) exists, we have defined a 25% relative change as the MCID for this outcome.
Quality of life (QoL) assessed with: EQ-5D-3L (lower score is worse) follow-up: 90 days	-	SMD 0.14 lower (0.49 lower to 0.22 higher)	-	3165 (3 RCTs)	⊕⊕⊕⊖ Moderate ^{b,d,e}	Intensive systolic blood pressure management likely results in little to no clinically meaningful difference in quality of life. Since no established MCID exists, we have defined a 0.5 SD change as the MCID for this outcome.
Neurologic adverse events assessed with: any type of intracranial hemorrhage follow-up: range 1 days to 90 days	277 per 1000	274 per 1000 (241 to 316)	RR 0.99 (0.87 to 1.14)	4304 (9 RCTs)	⊕⊕⊕⊕ High ^{a,c,f}	Intensive systolic blood pressure management does not increase or reduce neurologic adverse events. Since no established MCID exists, we have defined a 25% relative change as the MCID for this outcome.

Neurologic adverse events (symptomatic) assessed with: symptomatic intracranial hemorrhage follow-up: range 1 days to 90 days	45 per 1000	47 per 1000 (35 to 62)	RR 1.03 (0.77 to 1.36)	4304 (9 RCTs)	⊕⊕⊕⊖ Low ^{a,f,g}	Intensive systolic blood pressure management may result in little to no clinically meaningful difference in symptomatic neurologic adverse events. Since no established MCID exists, we have defined an absolute change of 1% as the MCID for this outcome.
Neurologic status assessed with: NIHSS (higher score is worse) Scale from: 0 to 42 follow-up: range 1 days to 10 days	The mean neurologic status was 0	MD 1.77 higher (0.37 higher to 3.16 higher)	-	1710 (6 RCTs)	⊕⊕⊕⊖ Moderate ^{b,e,h}	Intensive systolic blood pressure management probably results in little to no clinically meaningful difference in neurologic status (continuous). Since no established MCID exists, we have defined a 0.5 SD change as the MCID for this outcome.
Neurologic status (dichotomous) assessed with: NIHSS follow-up: range 1 days to 10 days	193 per 1000	137 per 1000 (98 to 195)	RR 0.71 (0.51 to 1.01)	713 (3 RCTs)	⊕⊕⊕⊖ Moderate ^{a,f,i}	Intensive systolic blood pressure management probably reduces the number of participants with a favorable neurologic status (dichotomous). Since no established MCID exists, we have defined a 25% relative change as the MCID for this outcome.
All-cause mortality assessed with: number of deaths follow-up: range 1 days to 90 days	114 per 1000	136 per 1000 (123 to 151)	RR 1.19 (1.08 to 1.32)	4297 (9 RCTs)	⊕⊕⊕⊖ Moderate ^{a,f,j}	Intensive systolic blood pressure management probably increases all-cause mortality. Since no established MCID exists, we have defined an absolute change of 1% as the MCID for this outcome.
Hospital length of stay assessed with: length of hospital stay in days follow-up: 90 days	The mean hospital length of stay was 0	MD 0.38 higher (1.4 lower to 2.15 higher)	-	3086 (5 RCTs)	⊕⊕⊕⊖ Moderate ^{b,e,k}	Intensive systolic blood pressure management probably results in little to no clinically meaningful difference in hospital length of stay. Since no established MCID exists, we have defined a 0.5 SD change as the MCID for this outcome.
Other adverse events assessed with: any non-neurological adverse event follow-up: range 1 days to 90 days	207 per 1000	251 per 1000 (178 to 357)	RR 1.21 (0.86 to 1.72)	4012 (8 RCTs)	⊕⊖⊖⊖ Very low ^{i,l,m}	Intensive systolic blood pressure management may have little to no clinically meaningful effect on other adverse events, but the evidence is very uncertain. Since no established MCID exists, we have defined a 25% relative change as the MCID for this outcome.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; MD: mean difference; RR: risk ratio; SMD: standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.grade.pro.org/presentations/#/isof/isof_question_revman_web_466055083624089443.

^a We did not downgrade the certainty of the evidence, as we judged most studies to have an overall low risk of bias for this outcome.

^b We did not downgrade the certainty of the evidence due to inconsistency, since the heterogeneity was explained by the type of procedure performed. See subgroup analysis.

^c We did not lower the certainty of the evidence due to imprecision, as the 95% confidence interval did not cross the significance threshold for benefit or harm, indicating no meaningful difference or effect. As no established MCID was identified in the literature, we defined a 25% relative change as the MCID for this outcome.

^d We downgraded one level due to risk of bias as all the included studies for this outcome had overall some concerns about risk of bias.

^e We did not lower the certainty of the evidence due to imprecision, as the 95% confidence interval did not cross the significance threshold for benefit or harm, indicating no meaningful difference or effect. As no established MCID was identified in the literature, we defined a 0.5 SD change as the MCID for this outcome.

^f We did not downgrade the certainty of the evidence due to inconsistency since the heterogeneity was not important ($\leq 30\%$).

^g We downgraded two levels due to imprecision, as the pre-established MCID of 1% for this outcome indicates that its 95% CI includes both minimal, clinically non-meaningful benefit and potential important harm.

^h We downgraded one level due to risk of bias because three of the six studies included for this outcome had a high overall risk of bias.

ⁱ We downgraded one level due to imprecision, as considering the prespecified MCID of 25% relative risk change, the 95% CI for the pooled estimate includes both no effect and a harmful MCID.

^j We downgraded one level due to imprecision, as considering the prespecified MCID of 1% absolute risk change, the 95% CI for the pooled estimate includes both no effect and a harmful MCID.

^k We downgraded one level due to risk of bias since three of five included studies presented an overall high risk or some concerns about risk of bias.

^l We downgraded one level due to risk of bias, since half of the studies presented an overall high risk or some concerns about risk of bias.

^m We downgraded the certainty of the evidence by one level due to inconsistency, as the heterogeneity was substantial and lacked a clear explanation.

BACKGROUND

Description of the condition

An ischemic stroke (80% to 85% of all strokes) is caused by the disruption of blood flow, leading to brain tissue damage and functional loss, with subsequent impact on individuals' life and functionality [1, 2]. According to the Global Burden of Disease Study conducted in 2021, stroke is the third leading cause of combined death and disability (measured in disability-adjusted life years) and the second leading cause of death worldwide [3, 4]. With the aging of the population, strokes have become a significant contributor to the personal and financial burdens on society [5]. The direct and indirect costs of strokes in the US were estimated at 52.8 billion US dollars annually from 2017 to 2018 [6]. Ischemic stroke is usually caused by the thrombotic occlusion of a cerebral artery, with therapeutic alternatives in the acute period including intravenous thrombolytic treatment with recombinant tissue plasminogen activator, intra-arterial fibrinolysis, and endovascular thrombectomy [5, 7]. Thrombolytic drugs aim to dissolve the thrombus, thereby restoring blood flow. Endovascular thrombectomy involves recanalization therapies in which the blood clot is removed using a mechanical device. The ultimate goal is to eliminate the clot, achieving recanalization of the affected cerebral tissue. If the procedure is performed in a timely and uncomplicated manner, the neurologic functional outcome can improve significantly [8, 9]. Despite recanalization efforts, 50% to 60% of people undergoing endovascular therapy for a stroke died or remained disabled at 90 days, regardless of whether it was a primary or extended window intervention [10].

Description of the intervention and how it might work

The leading challenge in acute ischemic stroke research is currently beyond reperfusion and focuses on ischemia-reperfusion injuries, encompassing blood–brain barrier disruption, cerebral edema, and intraparenchymal hemorrhage [11]. Arterial hypertension stands as one of the most significant modifiable risk factors for stroke onset [12]. Nonetheless, the relationship between blood pressure at the time of stroke onset and the outcomes is complex. The impact of blood pressure on the outcomes of people with stroke is influenced by multiple factors, including baseline blood pressure, degree of collateral circulation, size of the ischemic penumbra area, and achieved recanalization, among others [12]. Managing blood pressure during and after a cerebral revascularization procedure is crucial. Sustained high blood pressure during and after thrombectomy or thrombolysis (or both) can worsen clinical outcomes due to bleeding and cerebral edema as a result of increased susceptibility of ischemic tissue to pressure variations, leading to microvascular integrity loss. However, treatment to reduce blood pressure may decrease perfusion pressure and consequently increase the size of the ischemic lesion [12, 13]. Given all this complexity, the optimal blood pressure management in people with reperfused ischemic stroke is uncertain [12].

People undergoing cerebral reperfusion typically receive postreperfusion care in stroke units or intensive care units equipped with continuous blood pressure monitoring using invasive or non-invasive devices [14, 15, 16]. Current guidelines recommend maintaining blood pressure below 180/105 mmHg for 24 hours after endovascular thrombectomy or systemic thrombolysis (or both) [15, 16], with no specific blood pressure-lowering agent recommended [16]. Options for managing arterial

hypertension include labetalol, nicardipine, and clevidipine [15]. However, data gathered from the StrokeNet survey showed that most US institutions use a systolic blood pressure target lower than 140 mmHg in reperfused patients [17]. Observational studies have reported that lower systolic blood pressure after successful endovascular thrombectomy and systemic thrombolysis may be associated with better functional outcomes [10]. Non-randomized studies have suggested an association between increased systolic blood pressure values and symptomatic intraparenchymal hemorrhage, as well as mortality, in the setting of successful reperfusion after therapy [17, 18]. Specifically, one prospective multisite observational study identified a maximal systolic blood pressure of 160 mmHg within 24 hours after endovascular thrombectomy, which distinguished between people with good versus poor functional outcomes [19]. Others showed that moderate systolic blood pressure levels between 140 mmHg and 160 mmHg were associated with better patient outcomes [18]. Nevertheless, one randomized trial conducted on Chinese people who underwent successful endovascular thrombectomy discovered that maintaining systolic blood pressure below 120 mmHg was harmful, with worse functional outcomes [20]. In recent years, several trials have been conducted to help clarify this area of uncertainty [10, 11, 14, 19, 20, 21].

Why it is important to do this review

Current recommendations in the guidelines for ischemic stroke management suggest maintaining blood pressure below 180/105 mmHg for 24 hours after endovascular thrombectomy [15, 16]. Similar blood pressure values are recommended in cases of systemic thrombolysis. However, the evidence supporting these recommendations stems from non-randomized studies with low-certainty evidence [15, 16]. Some studies have suggested that achieving lower values of systolic blood pressure (reducing it to less than 160 mmHg) after the procedure would be beneficial in achieving the best functional outcome and reducing the risk of symptomatic intracranial hemorrhage [12, 19].

Considering the global incidence of ischemic strokes, the investments made in procedures for achieving cerebral reperfusion, the recognized efficacy of reperfusion as a treatment for ischemic stroke, and the crucial significance of blood pressure management during the acute phase, along with the existing uncertainties in current recommendations and recent clinical trials aiming to provide clarity, it becomes essential to establish the appropriate blood pressure management for this population. This is crucial to minimize the risk of hemorrhage and complications, confine the ischemic area, and thereby optimize the functional outcome, resulting in an improved quality of life for individuals and a reduced burden on individuals, families, healthcare systems, and society as a whole.

OBJECTIVES

To assess the benefits and harms of intensive systolic blood pressure management (target less than 160 mmHg) versus conventional management (target less than 180 mmHg) in people undergoing ischemic stroke reperfusion via systemic thrombolysis or endovascular thrombectomy.

METHODS

We followed the Methodological Expectations for Cochrane Intervention Reviews when conducting the review [22], and PRISMA 2020 for the reporting [23].

Differences between protocol and review

Originally, we included clinical function as a critical outcome and analyzed this both continuously and dichotomously. Considering the ordinal nature of the modified Rankin Scale, we retained the dichotomous analysis as a critical outcome, reclassified the continuous analysis as an important outcome, and removed it from [Summary of findings 1](#).

We analyzed neurologic adverse events and symptomatic neurologic adverse events separately, given how they were reported in the studies, to include all available data. We defined a minimal clinically important difference (MCID) of 1% absolute change for symptomatic neurologic adverse events and a 25% change in risk ratio (RR) as the MCID for neurologic adverse events [24].

We only analyzed all-cause mortality as a long-term outcome, as no information was reported for the acute period. We defined an MCID of 1% for the absolute change in this outcome [24]. Finally, for other adverse events, we defined an MCID as a 25% change in risk ratio.

We did not use the RoB 2 tool for cluster-RCTs because we did not include any.

We did not conduct a publication bias assessment, as we included fewer than 10 trials.

We could only conduct the sensitivity analyses excluding unpublished data for the 'clinical function continuous' outcome. We were unable to conduct a sensitivity analysis as per the study design, as all included studies were parallel RCTs.

We did not produce a summary of findings table based on the equity assessment, as we found no differences in subgroup tests by sex and socio-economic status in the available data for our critical outcomes.

Criteria for considering studies for this review

Types of studies

We included randomized controlled trials (RCTs) and cluster-RCTs. Following the guidelines in Chapter 23 of the *Cochrane Handbook for Systematic Reviews of Interventions* [25], we excluded cross-over trials, because they may have a potential withdrawal, rebound, or carry-over effect during or after the wash-out period. We excluded quasi-RCTs (controlled trials that used inappropriate strategies of allocating interventions) and non-randomized clinical trials, following the algorithm outlined in Chapter 24 of the *Cochrane Handbook*, due to the availability of RCTs that address our research questions [26].

Types of participants

We included adults (aged older than 18 years) with acute ischemic stroke (computed tomography or magnetic resonance scanning, having excluded intracranial hemorrhage before randomization) reperfused by systemic thrombolysis or

endovascular thrombectomy. Systemic thrombolysis could be performed with different drugs, doses, and administration routes (intravenous or intra-arterial) established by the studies, while endovascular thrombectomy could involve thromboaspiration, stent retriever thrombectomy, aspiration, or a combination of techniques. We included any post-procedure reperfusion status defined by the authors as inclusion criteria for the study.

If we identified studies in which only a subset of participants was relevant to this review, we included these studies if data were available separately for the relevant subset, or if more than 80% of participants met the inclusion criteria.

Types of interventions

We included studies comparing intensive systolic blood pressure management (target any systolic blood pressure objective less than 160 mmHg) versus conventional management (target systolic blood pressure less than 180 mmHg) in people who underwent systemic thrombolysis or endovascular thrombectomy for reperfusion of ischemic stroke.

If studies used different systolic blood pressure management cutoff points, we used the classification defined by the study authors.

We included co-interventions, provided they were not part of the randomized treatment, and were comparable in both the intervention and comparator groups, to establish a fair comparison. If a study included multiple arms, we included any arm that met the inclusion criteria for this review.

Interventions

Intensive systolic blood pressure management (target any systolic blood pressure objective less than 160 mmHg) for at least 24 hours after the therapeutic procedure (systemic thrombolysis or endovascular thrombectomy), monitored through invasive or non-invasive methods and utilizing the recommended intravenous antihypertensive medications as defined by study authors (usually nicardipine, although the use of other antihypertensive agents such as labetalol, hydralazine, etc. could be allowed). According to the recommendations from the European and American guidelines, the minimum duration of the intervention was 24 hours post-procedure [15, 16], with randomization and blood pressure management initiated within the first hour after the procedure or according to the study authors' specifications.

The systolic blood pressure cutoff of less than 160 mmHg was chosen as it was below current guideline recommendations, with the aim of including the greatest number of RCTs comparing targets below the current guideline thresholds [15, 16].

Comparators

Conventional systolic blood pressure management (target systolic blood pressure less than 180 mmHg) for at least 24 hours after the therapeutic procedure (systemic thrombolysis or endovascular thrombectomy), monitored through invasive or non-invasive methods and utilizing the recommended intravenous antihypertensive medications as defined by study authors (usually nicardipine, although the use of other antihypertensive agents such as labetalol, hydralazine, etc. could be allowed). According to the recommendations from the European and American guidelines, the minimum duration of the intervention was 24 hours post-procedure [15, 16], with randomization and blood pressure management

initiated within the first hour after the procedure or according to the study authors' specifications.

Outcome measures

We did not exclude a study if it failed to report more than one of our critical or important outcome measures of interest. We only excluded studies if none of our outcomes of interest were measured, and provided there was evidence to support this (e.g. contact with study authors, access to the original protocol, etc.).

Critical outcomes

- **Clinical function (dichotomous):** assessed at the end of the scheduled follow-up period, measured on the modified Rankin Scale (mRS) (a standard global seven-level measure of disability; score 0 to 2 indicating favorable functional outcome, score 3 to 6 indicating poor outcome, dependence, or death). Given that some studies used a definition of 'favorable outcome' as a score of 0 to 1, we also looked for data on the number of participants in each individual mRS category. If the mRS score was not reported, we used the study's definition of functional outcome. There was no established MCID, so we considered a change in risk ratio (RR) of 25% as the MCID for the dichotomous outcome [27].
- **Health-related quality of life:** assessed on a three-level EuroQoL 5-Dimension Self-Report Questionnaire (EQ-5D) or any other validated tool as defined by the study authors. The data were analyzed as continuous data, with the primary measure being the total score at follow-up. The MCID will range from 0.08 to 0.12 for EQ-5D [28]. We planned to calculate standardized mean differences (SMD) with 95% confidence intervals (CI) if data were pooled from studies using different instruments to measure quality of life. Since we used SMD, we considered a MCID of 0.5 standard deviations.
- **Neurologic adverse events:** assessed as the frequency of postprocedural intracerebral hemorrhage (including symptomatic intracranial hemorrhage or intracranial hemorrhage (or both) of any type in brain imaging within seven days of treatment or less) or malignant cerebral edema. We defined symptomatic intracranial hemorrhage according to the criteria used in the third European Cooperative Acute Stroke Study (ECASS III) [29]. When symptomatic intracranial hemorrhages were not reported according to these criteria, we used the study's definition. Note that symptomatic intracranial hemorrhage includes hemorrhagic transformation of the infarct, hemorrhage elsewhere in the brain remote from the infarct, and hemorrhage into the spaces surrounding the brain. Malignant cerebral edema refers to a condition characterized by rapid neurologic deterioration accompanied by substantial brain swelling seen on computer tomography or magnetic resonance imaging, which often leads to death or poor functional outcomes. We analyzed neurologic and symptomatic neurologic adverse events separately to include all data, defining a 1% absolute change as an MCID for symptomatic events and a 25% risk ratio change for neurologic events.

Important outcomes

- **Clinical function (continuous):** assessed at the end of the scheduled follow-up period, measured on the modified Rankin Scale (mRS). For the continuous outcomes' analysis, we considered an MCID of 0.5 standard deviations of the SMD [27].

- **Neurologic status:** assessed by NIHSS (National Institutes of Health Stroke Scale) scores at the end of follow-up. We defined a favorable neurologic outcome as an NIHSS score of 0 to 1. As no MCID is established, we considered a change in RR of 25% for dichotomous variables (e.g. good outcome). For continuous outcomes (e.g. each individual NIHSS score), we considered an MCID of 0.5 standard deviations of the SMD.
- **All-cause mortality:** assessed by the frequency of death from all causes following a stroke during the acute phase (within seven to 14 days) and at the end of scheduled follow-up (the established minimum clinical follow-up will be 90 days). We defined an MCID of 1% for the absolute change in this outcome.
- **Hospital length of stay:** assessed by median duration of hospitalization in days. As no MCID was established, we considered an MCID of 0.5 standard deviations of the SMD.
- **Other adverse events:** assessed by the frequency of other adverse events reported in the studies, such as heart failure, arterial hypotension, sepsis, arterial re-occlusion, aspirational pneumonia, pseudoaneurysm at the puncture site, non-cerebral hemorrhage, and other reported adverse events occurring at any point during the established follow-up. We considered an MCID of 25% change in risk ratio.

Timing of outcome measurement

We categorized outcomes into two sets of time points.

- Short-term postintervention (defined as zero to two weeks postintervention) to detect illness recovery/symptom reduction following the intervention and any adverse events [1].
- Long-term (defined as longer than two weeks postintervention) as a measure of medium- to long-term to detect illness recovery/symptom reduction [1].

When multiple results were reported for each outcome, we included the longest follow-up period in each category.

Search methods for identification of studies

Electronic searches

We searched the following sources from the beginning of each database to March 2025. We did not apply any restrictions on language, date, or publication status.

- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 3) in the Cochrane Library (searched March 2025).
- MEDLINE (Ovid MEDLINE ALL 1946 to March 2025).
- Embase (Ovid; 1974 to March 2025).
- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (www.clinicaltrials.gov; searched March 2025).
- World Health Organization International Clinical Trials Registry Platform (ICTRP) (www.who.int/trialsearch/; searched March 2025).

Two filters were applied in MEDLINE (Ovid): the Cochrane Highly Sensitive Search Strategy for Identifying Randomized Trials: Sensitivity-maximizing Version (2023 revision) and the Embase (Ovid) Cochrane Highly Sensitive Search Strategy for Identifying Randomized Trials (2023 revision) [30].

Details of the search strategies are available in [Supplementary material 1](#).

Searching other resources

We performed a search in *Web of Science – Conference Proceedings Citation Index–Science (CPCI-S)* from 1990 to May 2026 to find conference abstracts on thrombolysis in acute ischemic stroke.

We checked the reference lists of all included studies and relevant review articles. We contacted experts in the field to identify additional unpublished data.

Before submitting this publication, we checked for comments, corrections, or retractions in the included studies.

Data collection and analysis

Selection of studies

We used Covidence software for study selection [31]. Two review authors (LBV and SD) independently screened the titles and abstracts of references identified by the searches. Any disagreements between the authors were resolved with input from a third review author (LIG). We obtained full-text copies of all potentially relevant reports. The same two review authors (LBV and SD) independently screened the full-text studies and recorded reasons for the exclusion of excluded studies. Any disagreements between the authors were resolved with input from a third review author (LIG) if needed. If we could not resolve a disagreement, we categorized the study as awaiting classification and contacted the study authors for clarification. We presented a PRISMA flow diagram to show the process of study selection [23]. We listed all articles excluded after full-text assessment in the 'Characteristics of excluded studies' table and provided the reasons for exclusion [23].

Data extraction and management

For studies that fulfilled our inclusion criteria, two review authors (LBV and SD) independently extracted key information, using a piloted data extraction form, on participants, interventions, and comparators. The review authors were not blinded to journals or institutions. Any disagreements between the authors were resolved by discussion, with input from a third review author (LIG) if needed.

One review author (LBV) entered the data into Review Manager (RevMan) [32]. Another two review authors (LIG and SD) checked and transferred the study data into the analyses.

We extracted the following data from the reports.

- Methods
 - Study design: study dates (start date to end date; if dates were not available, we reported this), study settings, country, language of publication, and study identifier.
- Participants
 - Inclusion and exclusion criteria.
 - Participant details, baseline demographics (mean age, age range, gender, residential location, diagnosis criteria used for acute ischemic stroke, severity of the condition, location of the arterial occlusion, time interval from stroke onset to randomization, treatment performed to achieve reperfusion in stroke, time to actual delivery of therapy/recanalization, modality of percutaneous vascular intervention or agent and dose of pharmacologic thrombolytic therapy used, early ischemic changes on computer tomography according to the Alberta Stroke Program Early CT Score (ASPECTS), inclusion criteria and exclusion criteria).

- Number of participants in each treatment group with outcome events.
- Interventions and comparisons, according to the Template for Intervention Description and Replication (TIDieR) checklist [33, 34]:
 - Name of the intervention.
 - Why: rationale, theory, or goal of the elements essential to the intervention.
 - What: physical or informational materials used in the intervention, procedures, activities, or processes used in the intervention.
 - Who provided: expertise, background and specific training given.
 - How: describe modes of delivery.
 - Where: describe the location where the intervention occurred, including infrastructure and features.
 - When/how much: the number of times the intervention was delivered over a period of time.
 - Tailoring: describe if personalization or adaptations were planned.
 - Modifications: during the course of the study.
 - How well: measurements of adherence or fidelity.
- Outcomes
 - Definitions of relevant outcomes, method and timing of outcome measurement, as well as any subgroups relevant to the review.
- Study funding sources.
- Declarations of interest, by primary investigators.

We reported these data in the 'Characteristics of included studies' table and summarized them in the results section. We contacted the authors of the included studies to inquire whether they were willing to answer questions regarding their studies. We documented these communications. We sought relevant missing information on the study from the primary study authors if required.

Dealing with duplicate and companion publications

In the event of duplicate publications, companion documents, or multiple reports of a primary study, we maximized the information yield by collating all available data, and we used the most complete data set, aggregated across all known publications. We listed duplicate publications, companion documents, multiple reports of a primary study, and trial documents of included trials (such as trial registry information) as secondary references under the study ID of the included study. We also listed duplicate publications, companion documents, multiple reports of a study, and trial documents of excluded trials (such as trial registry information) as secondary references under the study ID of the excluded study.

Data from clinical trial registers

If data from included studies were available as study results in clinical trial registers, such as ClinicalTrials.gov or similar sources, we made full use of this information and extracted the data. Where there was also a full publication of the study, we collated and critically appraised all available data. If the published and unpublished data did not match, we asked the study authors for clarification. If we received no response, we presented the discrepancies in the review. If an included study was labeled as completed in a clinical trial register but no additional information

(study results, publication, or both) was available, we added this study to the 'Characteristics of studies awaiting classification' table.

Risk of bias assessment in included studies

Two review authors (LBV and SD) independently assessed the risk of bias for the results of the main outcomes using the Cochrane RoB 2 tool for all outcomes at the latest follow-up [35, 36, 37]. For all outcomes, we analyzed data using the intention-to-treat principle [36, 38]. We resolved disagreements by consulting a third review author (LIG). If inadequate information was unavailable from the publications, trial protocols, clinical study reports, or other sources, we contacted the study authors for more details on the risk of bias items.

We assessed the risk of bias for all RoB 2 domains and judged each domain as 'high risk of bias,' 'some concerns,' or 'low risk of bias' using the responses to the signaling questions and algorithms within the RoB 2 tool. Domains included bias caused by the randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported results. We used the tool algorithm to reach an overall risk of bias judgment for each outcome. We quoted evidence to support our judgments in the risk of bias table and, if we disagreed with a judgment recommended by the algorithm, we included an explicit statement as to why. We managed our risk of bias assessments using the RoB 2 Excel tool [37]. All data are publicly available, as supplementary material, in a public repository.

For cluster-RCTs, we would have used the RoB 2 tool and an additional domain specifically for cluster-RCTs from the archived version of the tool (domain 1b – "bias arising from the timing of identification and recruitment of participants"), available at www.riskofbias.info, with its corresponding signaling questions. We would have followed the guidance in the *Cochrane Handbook* (Section 23.1.2 and Table 23.1.a) [25].

Measures of treatment effect

We expressed dichotomous data (e.g. clinical functional outcome as a favorable functional outcome and a poor functional outcome) as a RR with 95% CIs. For continuous outcomes (e.g. neurologic status assessed by NIHSS scores measured on the same scale), we estimated the intervention effect using the mean difference (MD) with 95% CIs. When data were pooled from studies that used different instruments to measure the same outcome, we planned to calculate SMDs with 95% CIs. We entered data presented as a scale with a consistent direction of effect, and multiplied the SMD by a standard deviation that was representative of the pooled studies (e.g. the standard deviation from a well-known scale used by several of the studies included in the analysis on which the result was based). We undertook meta-analyses only when it was meaningful (i.e. when treatments, participants, and the underlying clinical questions were similar enough for pooling to make sense). We described skewed data reported as medians and interquartile ranges narratively.

Unit of analysis issues

We took into account the level at which randomization occurred, and multiple observations for the same outcome. If more than one comparison from the same study was eligible for inclusion in the same meta-analysis, we either combined groups to create a single

pair-wise comparison, or appropriately reduced the sample size so that the same participants did not contribute data to the meta-analysis more than once (splitting the shared group into two or more groups). Although the latter approach offered some solutions for adjusting the precision of the comparison, it did not account for correlation arising from the inclusion of the same set of participants in multiple comparisons [38, 39]. The unit of analysis was the participants with reperfused acute ischemic stroke by thrombolysis or thrombectomy.

For cluster-RCTs, we would have considered the cluster as the unit of analysis, not the individual participants, in order to avoid unit of analysis errors, as stated in Section 23.1.1 of the *Cochrane Handbook* [25]. If the effect measure for the cluster had not been determined by appropriate methods in the included studies, we would have multiplied the standard error of the effect estimate (from an analysis ignoring clustering) by the square root of the design effect, calculated using an intraclass (or intracluster) correlation coefficient (ICC) of 0.02, following guidance in Sections 23.1.4 and 23.1.5 of the *Cochrane Handbook* [25].

Dealing with missing data

If possible, we obtained missing data from the authors of the included studies. We carefully evaluated important numerical data, such as screening; randomly assigned participants; and intention-to-treat, as-treated, and per-protocol populations in our risk of bias assessments. For this, we investigated attrition rates (e.g. dropouts, losses to follow-up, and withdrawals), and critically appraised issues concerning missing data and the use of imputation methods (e.g. last observation carried forward). We analyzed available data only. When this was not possible, and when missing data were deemed to introduce significant bias, we explored the impact of including such studies in the overall assessment of results through a sensitivity analysis.

Reporting bias assessment

We undertook extensive literature searches without restrictions on publication date or language in order to limit reporting bias. We used study protocols and trial registrations to assess studies for selective reporting. If we included 10 studies or more per outcome, we planned to use funnel plots to assess small-study effects. Several explanations might account for funnel plot asymmetry, including true heterogeneity of effect with respect to study size, poor methodologic design (and hence the bias of small studies), and selective non-reporting [40]. Therefore, we planned to interpret the results carefully [41]. We did not perform statistical analysis for funnel plot asymmetry as these tests typically have low power, following guidance in Chapter 13 of the *Cochrane Handbook* [42].

Synthesis methods

We completed all syntheses using Review Manager [32]. We planned to undertake a meta-analysis only if we judged that the participants, interventions, comparisons, and outcomes were sufficiently similar to ensure a clinically meaningful result. In the primary analysis, we included all studies regardless of their risk of bias.

We used a random-effects model given the expected clinical diversity. We chose the Restricted Maximum Likelihood (REML) estimator to estimate the between-trial variance. We used the

Hartung-Knapp-Sidik-Jonkman method to calculate a confidence interval for the meta-analysis effect estimate when at least three studies were available, and the estimate of heterogeneity was greater than 0. In other scenarios, we used the Wald-type method [43]. We interpreted random-effects meta-analyses with consideration of the whole distribution of effects, and present CIs. We performed statistical analyses according to the statistical guidelines presented in Chapter 10 of the *Cochrane Handbook* [44].

If meta-analysis was not possible, we summarized the results using a narrative synthesis, according to the Synthesis Without Meta-analysis guidelines, instead of a pooled statistical synthesis. We followed Chapter 12 of the *Cochrane Handbook* [45].

We performed subgroup analyses using the methodology described by Deeks and colleagues, as recommended in Section 10.11.3 of the *Cochrane Handbook* [44].

Investigation of heterogeneity and subgroup analysis

We visually examined the variability in point estimates and the overlap in CIs. We used the I^2 statistic to estimate the degree of heterogeneity present among the trials in each analysis [46]. If we identified substantial unexplained heterogeneity, we reported it and explored possible causes by prespecified subgroup analysis. We used the following approximate guide to interpretation, outlined in Chapter 10 of the *Cochrane Handbook* [44].

- 0% to 40%: might not be important
- 30% to 60%: may represent moderate heterogeneity
- 50% to 90%: may represent substantial heterogeneity
- 75% to 100%: considerable heterogeneity

We avoided the use of absolute cutoff values, but interpreted the I^2 statistic in relation to the size and direction of effects, and the strength of evidence for heterogeneity (e.g. P value from the χ^2 test, or CI for the I^2 statistic).

We expected the following characteristics to introduce clinical heterogeneity, and we planned to conduct subgroup analyses for these, including an investigation of interactions [47].

- Subgroups of intervention based on target systolic blood pressure levels: below 160 mmHg, below 140 mmHg, and below 120 mmHg.
- Gender/sex: men versus women, considering trials conducted on one gender exclusively or with over 80% of participants representing a specific gender to establish the subgroups.
- Socioeconomic status based on country income level: low-, middle-, and high-income countries, based on the World Bank classification (available at the World Bank website).
- Stroke severity: according to the NIHSS score (less than 17 points versus 17 points or greater, range 0 to 42 points, with higher scores indicating greater severity), or using each trial's cutoff for severe stroke.
- Procedure performed: systemic thrombolysis or endovascular thrombectomy, or both (type).

We used the formal test for subgroup interactions in Review Manager [32], acknowledging its limitations due to its observational nature and low power to detect differences with fewer than 10 studies per category [38, 44].

Sensitivity analysis

We planned to perform the following sensitivity analyses for the critical outcomes (clinical functional outcome, health-related quality of life, and neurologic adverse events) to explore the influence of the following factors on effect sizes, according to Section 10.14 of the *Cochrane Handbook* [44].

- Study design: excluding studies with cluster randomization
- Methodologic quality: removing studies with an overall high risk of bias
- Published data status: excluding unpublished data

Certainty of the evidence assessment

We presented the overall certainty of the evidence for each outcome specified below, according to the GRADE approach, which takes into account issues related to internal validity (overall risk of bias, inconsistency, imprecision, publication bias) and external validity (directness of results). Two review authors (LBV and SD) independently rated the certainty of the evidence for each outcome. We resolved any differences in assessment by discussion, or consulting a third review author (LIG).

We presented a summary of the evidence in a summary of findings table. This provided key information about the best estimate of the magnitude of effect, in relative terms and absolute differences, for each relevant comparison of alternative management strategies; the number of participants and studies addressing each important outcome; and a rating of overall confidence in effect estimates for each outcome. We created the summary of findings table using the methods described in the *Cochrane Handbook* [48], Review Manager, and GRADEpro GDT software [24, 32, 49].

We justified all decisions to downgrade the certainty of the evidence using informative footnotes and GRADE guidelines [50].

We created a summary of findings table that included all outcomes listed in the [Outcome measures](#) section for the comparison of intensive systolic blood pressure management versus conventional systolic blood pressure management, focusing on the longest follow-up and prioritizing continuous outcomes.

Equity considerations

We explored health inequity through two characteristics defined by PROGRESS-Plus: gender/sex and socioeconomic status [51]. See [Supplementary material 9](#) for additional details about study characteristics with equity-related implications.

The prevalence of stroke and its complications is higher in females, attributed to multiple factors including biologic, clinical, social, and healthcare system-related factors [3, 4, 6]. Consequently, we intended to conduct a subgroup analysis to assess whether the reviews consider the incorporation of different sexes.

Additionally, we analyzed the impact of residential location as, in 2019, the World Bank reported that low-income countries exhibited an age-standardized stroke-related mortality rate of 3.6 (95% uncertainty interval 3.5 to 3.8) times higher and an age-standardized stroke-related DALY rate of 3.7 (95% uncertainty interval 3.5 to 3.9) times higher than those of high-income countries. Considering this reason for inequity, we added

socioeconomic status based on residential location as a subgroup analysis.

In our subgroup analysis, we aimed to conduct an equity-related assessment that considered the context in which the interventions and studies included were conducted. This approach allowed us to explore the implications of our findings across diverse populations. For the subgroup analysis, we extracted sex and residential location disaggregated data. Where these data were not reported, we contacted the study authors to obtain them.

We would have created a separate summary of findings table to describe the critical outcomes (clinical function, health-related quality of life, and neurologic adverse events) based on health inequity if we had found significant subgroup differences. Additionally, in the 'Equity-related implications for practice' and 'Equity-related implications for research' sections of the conclusions, we addressed the applicability of the results to different populations analyzed in our subgroups, highlighting potential disparities and context-specific factors that may influence outcomes and practice applicability.

Consumer involvement

We did not involve consumers in this review due to limited resources.

RESULTS

Description of studies

For details of the included studies, see [Supplementary material 2](#).

A complete record of all comparisons and analyses is available for this review in [Supplementary material 7](#), and a full data package is available in [Supplementary material 8](#).

Results of the search

We retrieved 7643 records from the literature search, and 6850 remained after duplicate removal (we automatically removed 767 records and manually removed 26). We excluded 6702 records after title and abstract screening. We sought the full texts of the remaining 148 records. We excluded 22 records, mainly due to ineligible populations and ineligible study design (see [Supplementary material 3](#) for detailed reasons for exclusion). We added four studies (six records) to the list of studies awaiting classification (see [Supplementary material 4](#)). We tried to contact the authors of these studies, but we received no responses. We found two ongoing studies (five records) (see [Supplementary material 5](#)).

We included nine studies with 115 references. Our screening of the reference lists of the included publications did not reveal any additional RCTs.

[Figure 1](#) shows the study flow diagram of the selection process. Eight studies were reported in English and one in Russian (Avidzba 2024 [52, 53, 54]). All included studies were published as full papers between 2019 and 2026. All studies had multiple reports that we could track and merge as unique studies.

Figure 1. PRISMA diagram

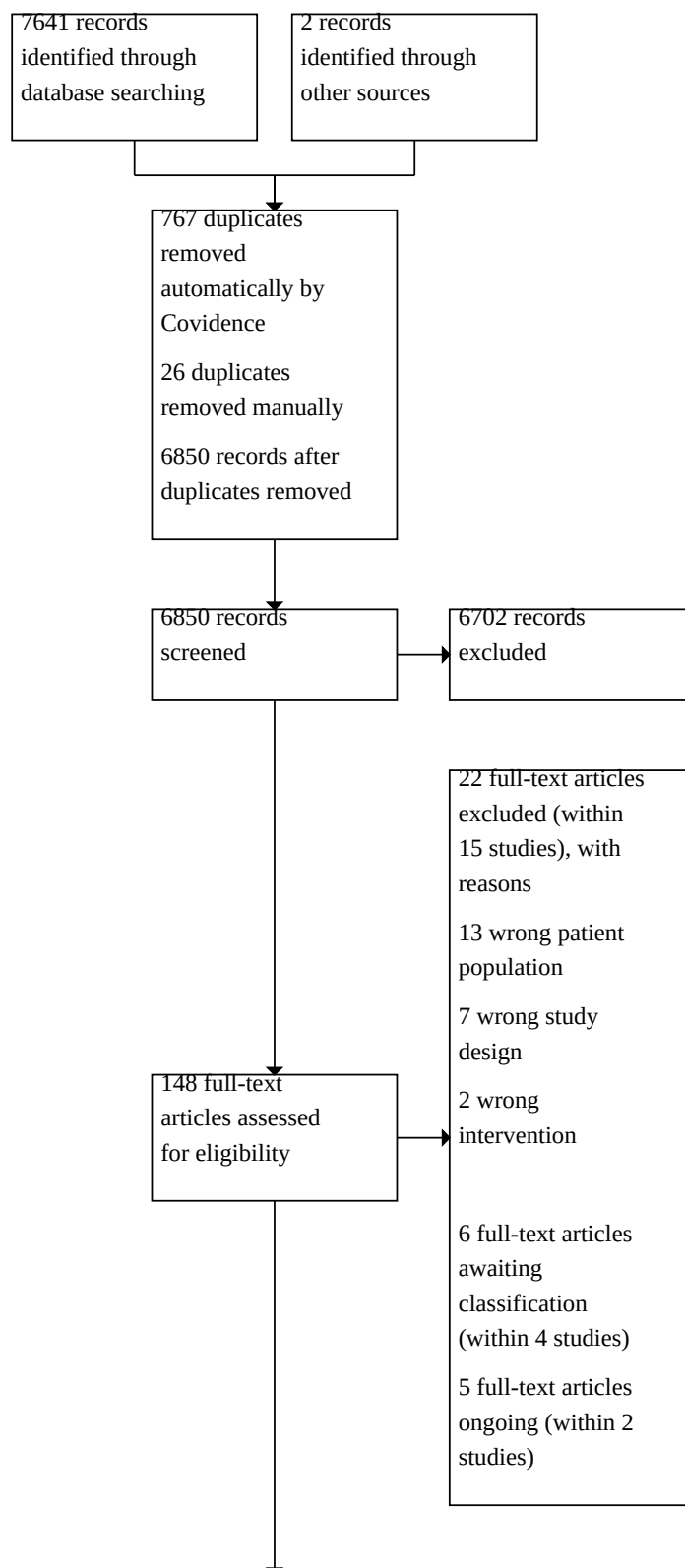
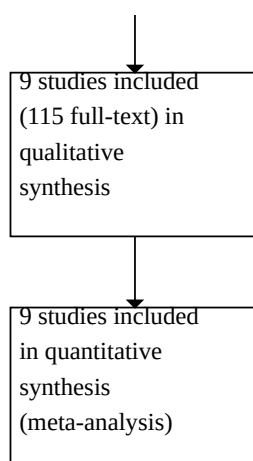


Figure 1. (Continued)



Included studies

Characteristics of study designs and settings

Table 1 and Table 2 summarize the included studies and synthesis. We included nine studies with a total of 4381 participants (Avidzba 2024; BP-TARGET [55, 56, 57, 58, 59, 60, 61, 62, 63]; CLEVER [64, 65]; DETECT [66, 67, 68]; ENCHANTED [69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143]; ENCHANTED2/MT [144, 145, 146, 147, 148, 149, 150, 151, 152]; IDENTIFY [153, 154, 155, 156]; OPTIMAL-BP [157, 158, 159, 160, 161, 162, 163, 164]; The BEST-II [165, 166]). All studies were RCTs, with a median sample size of 306 participants (interquartile range 85 to 602). The largest sample size was 2227 (ENCHANTED), and the smallest sample size was 30 (DETECT).

All studies were conducted between 2012 and 2024, in hospitalized patients. Three of the studies were single-center, and performed in the Russian Federation (Avidzba 2024), Canada (DETECT), and the United States (CLEVER). The other six studies were multicenter, five of them at a national level in China (ENCHANTED2/MT; IDENTIFY), Korea (OPTIMAL-BP), the United States (The BEST-II), and France (BP-TARGET). One study was multicenter and multinational (ENCHANTED) including patients from 15 countries (Australia, Brazil, Chile, China, Colombia, Hong Kong, India, Italy, South Korea, Singapore, Spain, Taiwan, Thailand, United Kingdom, Vietnam).

Characteristics of participants

All studies included adult participants. The mean age ranged from 66.7 to 85 years, and the percentage of female participants ranged from 37% to 70%. Seven of the studies included participants who underwent endovascular thrombectomy (BP-TARGET; CLEVER; DETECT; ENCHANTED2/MT; IDENTIFY; OPTIMAL-BP; The BEST-II) and the other two reperused with systemic thrombolysis (Avidzba 2024; ENCHANTED). However, most studies allowed participants to undergo other reperfusion treatments if indicated (prior systemic thrombolysis in studies focused on endovascular thrombectomy,

and vice versa). An overview of the characteristics of the included participants is given in Table 1 and Table 2.

Characteristics of interventions

All included studies employed intensive blood pressure management. Five studies used a target blood pressure of < 140 mmHg (BP-TARGET; DETECT; ENCHANTED; IDENTIFY; OPTIMAL-BP), two studies targeted < 120 mmHg (CLEVER; ENCHANTED2/MT), and one study targeted < 160 mmHg (Avidzba 2024). One study included two intervention arms with different targets (< 140 mmHg and < 160 mmHg) (The BEST-II). To achieve this blood pressure target, all studies used intravenous antihypertensive drugs when necessary, including urapidil, nicardipine, labetalol, hydralazine, azametonium, clevidipine, or other intravenous medicines according to local protocols. Only one study allowed the use of oral or topical antihypertensive drugs (ENCHANTED).

In nine studies, the randomized participants had to achieve the target blood pressure within the first hour after randomization and maintain it for at least 24 hours (Avidzba 2024; BP-TARGET; CLEVER; IDENTIFY; OPTIMAL-BP; The BEST-II), 48 hours (DETECT), or at least 72 hours (ENCHANTED; ENCHANTED2/MT).

The main comparison was conventional blood pressure management with a targeted systolic blood pressure of < 180 mmHg in eight studies (Avidzba 2024; BP-TARGET; DETECT; ENCHANTED; ENCHANTED2/MT; IDENTIFY; OPTIMAL-BP; The BEST-II) and < 160 mmHg in one study (CLEVER). If required, the study's investigator could prescribe an intravenous antihypertensive to achieve the blood pressure goal.

In general, the use of antihypertensive drugs was higher in the intervention group. None of the studies included co-interventions.

Outcomes

For our critical outcomes, all studies reported the clinical function dichotomously. Only three studies reported quality of life (ENCHANTED; ENCHANTED2/MT; OPTIMAL-BP), and all studies reported neurological adverse events. As all the studies reported the data for this outcome as any type of intracranial hemorrhage

(ICH) and symptomatic ICH, we extracted the data separately (Analysis 1.3; Analysis 1.4).

Regarding our important outcomes, only two studies reported clinical function as a continuous outcome (Avidzba 2024; DETECT). Additionally, the authors of another two studies provided us with these data (IDENTIFY; OPTIMAL-BP). Five studies reported hospital length of stay (Avidzba 2024; CLEVER; DETECT; ENCHANTED; ENCHANTED2/MT). The least reported outcome was neurologic status (as a dichotomous outcome), reported by just OPTIMAL-BP. Additionally, two study authors provided us with the data for this outcome (DETECT; IDENTIFY). Furthermore, neurological status as a continuous outcome was reported in five studies (Avidzba 2024; DETECT; ENCHANTED2/MT; OPTIMAL-BP; The BEST-II), and the authors of IDENTIFY sent us the data for the meta-analysis. Eight studies reported other adverse events (Avidzba 2024; BP-TARGET; CLEVER; DETECT; ENCHANTED; ENCHANTED2/MT; IDENTIFY; The BEST-II), and all studies reported all-cause mortality.

All nine studies had a long-term follow-up of 12 weeks or three months. Clinical function (dichotomous), quality of life, clinical function (continuous), all-cause mortality, and other adverse events were reported in long-term follow-up. On the other hand, neurological adverse events, neurological adverse events (symptomatic), hospital length of stay, neurological status, and neurological status (dichotomous) were measured and reported as part of the short-term follow-up.

Funding sources and conflicts of interest

All the studies received funding from some source. The primary funding source for most of the studies came from grants from foundations, universities, and other public governmental institutions. Three studies reported that the funder was not involved in the planning, conduct, analysis, and dissemination of the results (CLEVER; DETECT; ENCHANTED).

All the studies reported the presence/absence of conflicts of interest. One study declared no conflicts of interest (DETECT), and another one reported only a limited editorial role for one author, with no involvement in the publication decision and no other conflicts declared (Avidzba 2024). The remaining seven included studies received funding from governmental, institutional, academic, or non-profit sources. One study reported academic and governmental research grants, without clear industry-related conflicts in the information reported (IDENTIFY). The other six studies reported industry-related funding or author conflicts of interest, such as consulting fees, advisory board participation, speaker fees, research grants, patents or licensing royalties, stock options, or equity (BP-TARGET; CLEVER; ENCHANTED; ENCHANTED2/MT; OPTIMAL-BP; The BEST-II).

Ongoing studies and studies awaiting classification

We identified a total of two ongoing studies (ENCHANTED3/MT [167]; HOPE STUDY [168, 169, 170, 171]) from our broad literature search (see [Results of the search](#)), both exploring the effects of intensive versus conventional blood pressure management. The estimated completion date for the ENCHANTED3/MT is May 2026. The HOPE STUDY was listed as having unknown status on clinicaltrials.gov, so we contacted the author, who clarified that it is ongoing with an estimated completion date in mid-2026.

Four studies on the effect of intensive blood pressure management were classified as awaiting classification due to unknown status (CHASE-MT [172]; CRISIS I [173]; Sharma 2018 [174, 175, 176]) or because the full text was not available (Zhou 2017 [177]). We tried to contact the authors in May and June 2025 to clarify the studies' status, but received no response. We did not contact the authors of CRISIS I as there was no contact author information available.

See also [Supplementary material 4](#) and [Supplementary material 5](#) for more details about the ongoing and awaiting classification studies.

Excluded studies

We excluded 15 studies. Reasons for exclusion are documented in [Supplementary material 3](#).

Risk of bias in included studies

Risk of bias assessments for each outcome are located in the risk of bias table section within [Supplementary material 6](#) and at the side of the forest plots. To access further detailed risk of bias assessment data, visit <https://osf.io/dtp9f/overview>.

In general, all the included studies had an overall low risk of bias although, as an exception, three studies presented overall some concerns or high risk of bias for most of the different outcomes evaluated (Avidzba 2024; CLEVER; DETECT).

Two studies had an overall high risk of bias for clinical function as a dichotomous outcome, due to high risk arising from deviations from intended interventions (Avidzba 2024) and due to missing outcome data (DETECT). One study presented overall some concerns due to bias in the randomization process and deviation from intended interventions (CLEVER). The other six studies that evaluated this outcome had an overall low risk of bias (BP-TARGET; ENCHANTED; ENCHANTED2/MT; IDENTIFY; OPTIMAL-BP; The BEST-II).

Regarding quality of life, the three studies that evaluated it had some concerns, mainly due to bias in the measurement of the outcome (ENCHANTED; ENCHANTED2/MT; OPTIMAL-BP) and in the selection of the reported result (ENCHANTED). For our last critical outcome, neurological adverse events (symptomatic), two studies had an overall high risk of bias due to deviations from intended interventions (Avidzba 2024), and missing outcome data (DETECT), and one study had some concerns due to bias in the randomization process and deviation from intended interventions (CLEVER). For neurological adverse events (any ICH), three studies presented an overall high risk of bias due to deviations from intended interventions (Avidzba 2024), missing outcome data (DETECT), and selection of the reported result (OPTIMAL-BP). One study presented overall some concerns (CLEVER) and the other studies presented an overall low risk of bias for these outcomes.

Two out of four studies that evaluated clinical function as a continuous outcome had an overall high risk of bias (Avidzba 2024; DETECT), due to the high risk arising from deviations from intended interventions (Avidzba 2024) and due to missing outcome data (DETECT). The other two studies had an overall low risk of bias (IDENTIFY; OPTIMAL-BP).

In relation to the five studies that reported hospital length of stay as a continuous outcome, two of them had an overall high risk of bias

due to deviations from intended interventions (Avidzba 2024), and missing outcome data (DETECT), and one had some concerns due to bias in the randomization process and deviation from intended interventions (CLEVER). Overall, we judged a low risk of bias in the other two studies (ENCHANTED; ENCHANTED2/MT).

Three of six studies that evaluated neurologic status as a continuous outcome had an overall high risk of bias due to deviations from intended interventions (Avidzba 2024), missing outcome data (DETECT), and selection of the reported result (The BEST-II). We considered the other three studies to have an overall low risk bias (ENCHANTED; IDENTIFY; OPTIMAL-BP). For neurological status reported as a dichotomous outcome, three studies were included in the meta-analysis; one of them presented an overall high risk of bias due to missing outcome data (DETECT), while the others presented an overall low risk of bias (IDENTIFY; OPTIMAL-BP).

Two of the nine studies reporting all-cause mortality presented a high overall risk of bias due to deviations from intended interventions (Avidzba 2024), and missing outcome data (DETECT). One study presented overall some concerns due to bias in the randomization process and deviation from intended interventions (CLEVER). The other six studies presented an overall low risk of bias (BP-TARGET; ENCHANTED; ENCHANTED2/MT; IDENTIFY; OPTIMAL-BP; The BEST-II). Lastly, three of the eight studies reporting other adverse events presented a high risk of bias due to deviations from intended interventions (Avidzba 2024), missing outcome data (DETECT), and selection of the reported results (IDENTIFY). We

judged one study as having overall some concerns due to bias in the randomization process and deviation from intended interventions (CLEVER). We assessed the other four studies at overall low risk of bias (BP-TARGET; ENCHANTED; ENCHANTED2/MT; The BEST-II).

Synthesis of results

We summarized the absolute and relative effects for all critical and important outcomes, with their GRADE ratings signifying confidence in the effect estimates, in [Summary of findings 1](#). Additional results can be found in [Supplementary material 7](#).

1. Conventional versus intensive blood pressure management

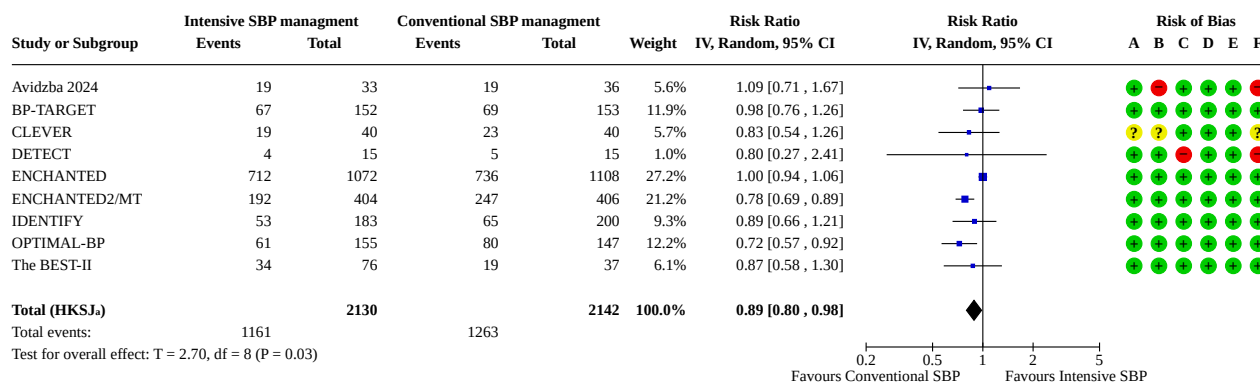
We included nine studies with a total of 4381 participants in this comparison.

Critical outcomes

Clinical function dichotomous

All nine studies reported clinical function dichotomously (Analysis 1.1) as the number of participants with an mRS score of 0 to 2 at 90 days (Avidzba 2024; BP-TARGET; CLEVER; DETECT; ENCHANTED; ENCHANTED2/MT; IDENTIFY; OPTIMAL-BP; The BEST-II). Intensive systolic blood pressure management results in little to no clinically meaningful difference in clinical function (dichotomous outcome) compared with conventional systolic blood pressure management (RR 0.89, 95% CI 0.80 to 0.98; $I^2 = 51\%$; 9 studies, 4272 participants; high-certainty evidence; [Figure 2](#)).

Figure 2. Forest plot for the outcome clinical function (dichotomous). Intensive versus conventional systolic blood pressure management.



Footnotes

^aCI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

^b Tau^2 calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Quality of life

Only three studies reported this outcome, including a total of 3165 participants (ENCHANTED; ENCHANTED2/MT; OPTIMAL-BP)

(Analysis 1.2). Intensive systolic blood pressure management likely results in little to no clinically meaningful difference in quality of life (SMD -0.14 , 95% CI -0.49 to 0.22 ; $I^2 = 75\%$; 3 studies, 3165 participants; moderate-certainty evidence; [Figure 3](#)).

Study or Subgroup	SMD	SE	Intensive SBP management	Conventional SBP management	Weight	Std. Mean Difference	Std. Mean Difference	Risk of Bias				
			Total	Total		IV, Random, 95% CI [points]	IV, Random, 95% CI [points]	A	B	C	D	E
ENCHANTED	0	0.04292	1068	1104	41.3%	0.00 [-0.08, 0.08]						
ENCHANTED2/MT	-0.20743	0.076457	342	346	33.6%	-0.21 [-0.36, -0.06]						
OPTIMAL-BP	-0.264074	0.11504	155	150	25.1%	-0.26 [-0.49, -0.04]						
Total (HKSJ)			1565	1600	100.0%	-0.14 [-0.49, 0.22]						
Test for overall effect: T = 1.66, df = 2 (P = 0.24)												
Heterogeneity: Tau ² (REML) = 0.02; Chi ² = 8.74, df = 2 (P = 0.01); I ² = 75%												

^bTau² calculated by Restricted Maximum-Likelihood method

(A) Bias arising from the randomization process
 (B) Bias due to deviations from intended interventions
 (C) Bias due to missing outcome data
 (D) Bias in measurement of the outcome
 (E) Bias in selection of the reported result
 (F) Overall bias

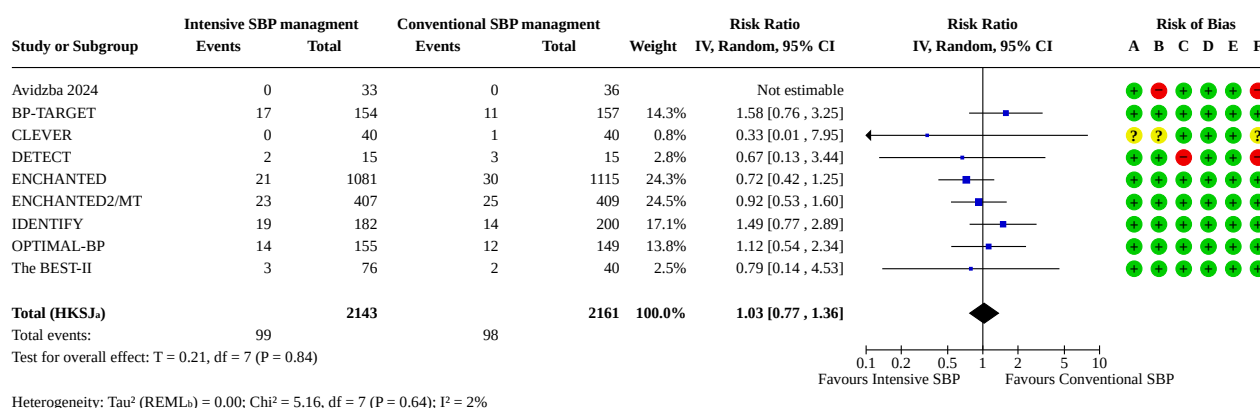
We analyzed neurologic adverse events and neurologic adverse events (symptomatic) separately due to the way they were reported in the included studies, with all the studies reporting this outcome as the number of participants who presented any type of ICH (Avidzba 2024; BP-TARGET; CLEVER; DETECT; ENCHANTED; ENCHANTED2/MT; IDENTIFY; OPTIMAL-BP; The BEST-II) (Analysis 1.3), and also as the number of participants with symptomatic ICH (Analysis 1.4) (neurologic adverse events (symptomatic)).

For any ICH, the total number of participants analyzed was 4304, showing that intensive systolic blood pressure management does not increase or reduce neurologic adverse events (RR 0.99, 95% CI 0.87 to 1.14; $I^2 = 30\%$; 9 studies, 4304 participants; high-certainty evidence; [Figure 4](#)). For symptomatic ICH, intensive systolic blood pressure management may result in little to no clinically meaningful difference in symptomatic neurologic adverse events in comparison with conventional systolic blood pressure (RR 1.03, 95% CI 0.77 to 1.36; $I^2 = 2\%$; 9 studies, 4304 participants; low-certainty evidence; [Figure 5](#)).

Study or Subgroup	Intensive SBP management		Conventional SBP management		Weight	Risk Ratio IV, Random, 95% CI	Risk Ratio IV, Random, 95% CI	Risk of Bias				
	Events	Total	Events	Total				A	B	C	D	E
Avidzba 2024	9	33	5	36	1.4%	1.96 [0.73, 5.26]						
BP-TARGET	67	158	69	160	14.4%	0.98 [0.76, 1.27]						
CLEVER	13	40	14	40	3.5%	0.93 [0.50, 1.72]						
DETECT	3	15	6	15	1.0%	0.50 [0.15, 1.64]						
ENCHANTED	160	1081	209	1115	20.4%	0.79 [0.65, 0.95]						
ENCHANTED2/MT	112	407	102	409	16.3%	1.10 [0.88, 1.39]						
IDENTIFY	103	179	103	196	21.0%	1.09 [0.91, 1.32]						
OPTIMAL-BP	83	155	78	149	18.0%	1.02 [0.83, 1.26]						
The BEST-II	26	76	12	40	4.0%	1.14 [0.65, 2.01]						
Total (HKSJ_a)		2144		2160	100.0%	0.99 [0.87, 1.14]						
Total events:	576		598									
Test for overall effect: T = 0.10, df = 8 (P = 0.92)												
Heterogeneity: Tau ² (REMI s) = 0.01; Chi ² = 11.06, df = 8 (P = 0.20); I ² = 30%												

^bTau² calculated by Restricted Maximum-Likelihood method.

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Figure 5. Forest plot for the outcome neurologic adverse events (symptomatic). Intensive versus conventional systolic blood pressure management.**Footnotes**

aCI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

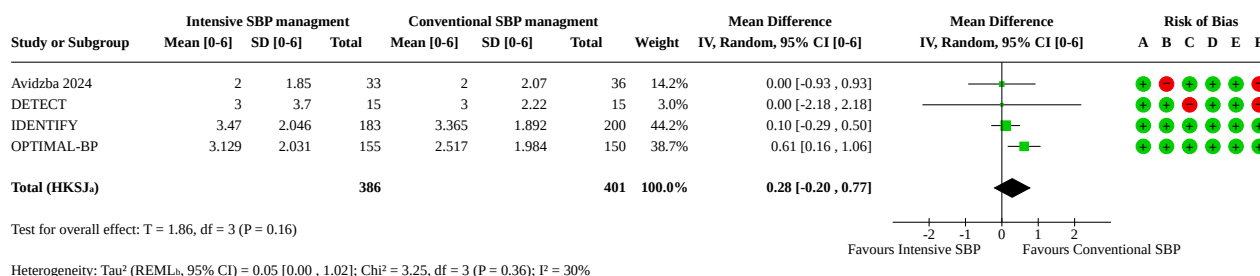
bTau² calculated by Restricted Maximum-Likelihood method.**Risk of bias legend**

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Important outcomes**Clinical function continuous**

Four studies with 787 participants studied clinical function as a continuous outcome (Avidzba 2024; DETECT; IDENTIFY; OPTIMAL-BP) (Analysis 1.5). Two of these studies reported it (Avidzba 2024; DETECT), while the authors of two other studies provided the data

for the analysis of this outcome continuously (IDENTIFY; OPTIMAL-BP). Intensive systolic blood pressure management likely results in little to no clinically meaningful difference in clinical function (continuous) compared to conventional systolic blood pressure management (MD 0.28, 95% CI -0.20 to 0.77; $I^2 = 30\%$; 4 studies, 787 participants; Figure 6).

Figure 6. Forest plot for the outcome clinical function (continuous). Intensive versus conventional systolic blood pressure management.**Footnotes**

aCI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

bTau² calculated by Restricted Maximum-Likelihood method.**Risk of bias legend**

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Neurologic status

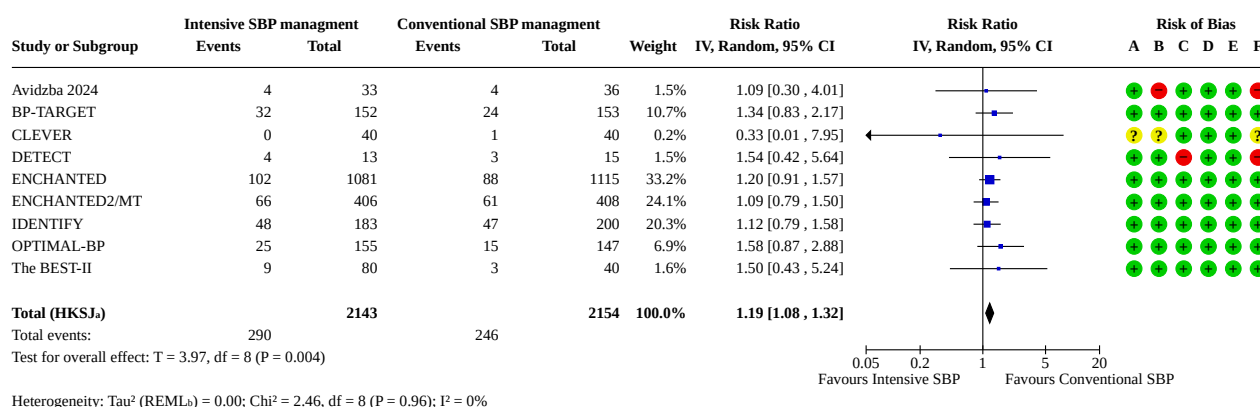
We meta-analyzed the data for this outcome as a continuous (Analysis 1.6) and as a dichotomous outcome (Analysis 1.7). For the first analysis, five studies reported the data (Avidzba 2024; DETECT; ENCHANTED2/MT; OPTIMAL-BP; The BEST-II), and the authors of one other study provided us with these data (IDENTIFY). Intensive systolic blood pressure management probably results in little to no clinically meaningful difference in neurologic status (continuous) (MD 1.77, 95% CI 0.37 to 3.16; $I^2 = 10\%$; 6 studies, 1710 participants; moderate-certainty evidence). Additionally, one study reported the NIHSS score dichotomously (OPTIMAL-BP), and the authors of two studies provided us with this data (DETECT; IDENTIFY). Intensive systolic blood pressure management probably reduces the number of participants with a favorable neurologic status (dichotomous), defined as a NIHSS score of 0 to 1 in the DETECT, IDENTIFY, and

OPTIMAL-BP studies or an improvement of more than 8 points in the OPTIMAL-BP study (RR 0.71, 95% CI 0.51 to 1.01; $I^2 = 0\%$; 3 studies, 713 participants; moderate-certainty evidence).

All-cause mortality

The nine studies, with 4297 participants, all studied this outcome (Avidzba 2024; BP-TARGET; CLEVER; DETECT; ENCHANTED; ENCHANTED2/MT; IDENTIFY; OPTIMAL-BP; The BEST-II) (Analysis 1.8). Only one study reported this outcome as in-hospital mortality (The BEST-II), while the other eight reported it during up to 90 days of follow-up. Intensive systolic blood pressure management probably increases all-cause mortality (RR 1.19, 95% CI 1.08 to 1.32; $I^2 = 0\%$; 9 studies, 4297 participants; moderate-certainty evidence; Figure 7).

Figure 7. Forest plot for the outcome all-cause of mortality. Intensive versus conventional systolic blood pressure management.



Footnotes

aCI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

bTau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Hospital length of stay

Five studies with 3086 participants studied and reported hospital length of stay (Avidzba 2024; CLEVER; DETECT; ENCHANTED; ENCHANTED2/MT). Intensive systolic blood pressure management probably results in little to no clinically meaningful difference in hospital length of stay (MD 0.38, 95% CI -1.40 to 2.15; $I^2 = 33\%$; 5 studies, 3086 participants; moderate-certainty evidence; Analysis 1.9).

Other adverse events

Eight studies, with 4012 participants, studied this outcome (Avidzba 2024; BP-TARGET; CLEVER; DETECT; ENCHANTED; ENCHANTED2/MT; IDENTIFY; The BEST-II) (Analysis 1.10). Intensive systolic blood pressure management may have little to no clinically meaningful effect on other adverse events, but the evidence is very uncertain (RR 1.21, 95% CI 0.86 to 1.72; $I^2 = 61\%$; 8 studies, 4012 participants; very low-certainty evidence).

2. Conventional versus intensive blood pressure management — subgroup analyses

The authors of studies kindly provided the data for the subgroup analysis by sex and stroke severity (Avidzba 2024; DETECT; IDENTIFY; OPTIMAL-BP), except for the ENCHANTED study, which reported disaggregated dichotomous data on clinical function. We also attempted to contact the authors of the remaining studies to request data for these subgroup analyses, but we received no response (BP-TARGET; ENCHANTED; ENCHANTED2/MT; The BEST-II). We conducted the other subgroup analyses (by country income level, intensive blood pressure target, and procedure performed) using the reported data. For more details, see analyses 2 to 6 in [Supplementary material 7](#).

The data should be interpreted with caution, and no further conclusions were drawn, due to the small number of trials (< 10) and participants (< 4381) in the subgroup analyses.

Critical outcomes

Clinical function (dichotomous)

For the subgroup analyses, see Analysis 2.1, Analysis 3.1, Analysis 4.1, Analysis 5.1, and Analysis 6.1 in [Supplementary material 7](#).

Investigating treatment effects between different country income levels, we found little to no difference in clinical function (dichotomous outcome) (test for subgroup differences: $\text{Chi}^2 = 1.13$, $\text{df} = 1$ ($P = 0.29$), $I^2 = 11.8\%$; Analysis 2.1).

In addition to that, we found a difference in the subgroup test according to the intensive blood pressure target in clinical function (dichotomous outcome) (test for subgroup differences: $\text{Chi}^2 = 10.15$, $\text{df} = 2$ ($P = 0.006$), $I^2 = 80.3\%$; Analysis 3.1). The intervention showed a less favorable effect as the blood pressure target became stricter (RR 0.78, 95% CI 0.64 to 0.96; 2 studies, 890 participants when targeting < 120 mmHg; RR 0.91, 95% CI 0.80 to 1.05; 6 studies, 3274 participants when targeting < 140 mmHg; and RR 0.98, 95% CI 0.20 to 4.76; 2 studies, 145 participants when targeting < 160 mmHg).

Regarding the subgroup analysis according to the type of procedure performed, we also found a difference in clinical function as a dichotomous outcome (test for subgroup differences: $\text{Chi}^2 = 27.69$, $\text{df} = 1$ ($P < 0.00001$), $I^2 = 96.4\%$; Analysis 4.1). The direction of the effect was more favorable to conventional blood pressure management than to intensive blood pressure management in the thrombectomy subgroup (RR 0.81, 95% CI 0.74 to 0.89; 7 studies, 2023 participants) versus the systemic thrombolysis group (RR 1.00, 95% CI 0.86 to 1.17; 2 studies, 2249 participants).

Lastly, in the subgroup analyses by sex and stroke severity, we also did not find subgroup differences in clinical function (dichotomous outcome) (respectively: test for subgroup differences: $\text{Chi}^2 = 0.18$, $\text{df} = 1$ ($P = 0.67$), $I^2 = 0\%$; Analysis 5.1; test for subgroup differences: $\text{Chi}^2 = 0.00$, $\text{df} = 1$ ($P = 0.96$), $I^2 = 0\%$; Analysis 6.1).

Quality of life

For the subgroup analyses, see Analysis 2.2, Analysis 3.2, and Analysis 4.2 in [Supplementary material 7](#).

We found little to no difference in quality of life in people according to the country where the study was conducted (test for subgroup differences: $\text{Chi}^2 = 0.17$, $\text{df} = 1$ ($P = 0.68$), $I^2 = 0\%$; Analysis 2.2).

Investigating treatment effects on quality of life between the three subgroups with the different intensive blood pressure targets showed no difference (test for subgroup differences: $\text{Chi}^2 = 0.41$, $\text{df} = 1$ ($P = 0.52$), $I^2 = 0\%$; Analysis 3.2).

For the subgroup analysis according to the procedure performed, we did find a subgroup difference (test for subgroup differences: $\text{Chi}^2 = 20.02$, $\text{df} = 1$ ($P < 0.00001$), $I^2 = 95.0\%$; Analysis 4.2) between the thrombectomy group (MD -0.22, 95% CI -0.56 to 0.11; 2 studies, 993 participants) versus the systemic thrombolysis group (MD 0, 95% CI -0.08 to 0.08; 1 study, 2172 participants).

We had disaggregated data from only one study for the subgroup analysis by sex and stroke severity for this outcome and found subgroup differences (test for subgroup differences by sex: $\text{Chi}^2 = 2.75$, $\text{df} = 1$ ($P = 0.10$), $I^2 = 63.7\%$; Analysis 5.2; by stroke severity $\text{Chi}^2 = 0.05$, $\text{df} = 1$ ($P = 0.82$), $I^2 = 0\%$; Analysis 6.2).

Neurologic adverse events

For the subgroup analyses, see Analysis 2.3, Analysis 2.4, Analysis 3.3, Analysis 3.4, Analysis 4.3, Analysis 4.4, Analysis 5.3, Analysis 5.4, Analysis 6.3, and Analysis 6.4 in [Supplementary material 7](#).

In the subgroup analysis for neurologic adverse events assessed as any ICH (symptomatic or not), we found no differences in the subgroup analysis according to country income level (test for subgroup differences: $\text{Chi}^2 = 1.60$, $\text{df} = 1$ ($P = 0.21$), $I^2 = 37.3\%$; Analysis 2.3). The same happened with neurologic adverse events (symptomatic) (Analysis 2.4); for this subgroup analysis by country income level, the results became less precise in both subgroups with no differences between them (test for subgroup differences: $\text{Chi}^2 = 0.02$, $\text{df} = 1$ ($P = 0.89$), $I^2 = 0\%$).

We also did not find a difference in the subgroup analysis according to the intensive blood pressure target for neurologic adverse events (test for subgroup differences: $\text{Chi}^2 = 2.51$, $\text{df} = 2$ ($P = 0.29$), $I^2 = 20.2\%$; Analysis 3.3) or neurologic adverse events (symptomatic) (test for subgroup differences: $\text{Chi}^2 = 1.01$, $\text{df} = 2$ ($P = 0.60$), $I^2 = 0\%$; Analysis 3.4).

The subgroup analysis according to the procedure performed did not show differences between subgroups for neurologic adverse events (any ICH, test for subgroup differences: $\text{Chi}^2 = 0.01$, $\text{df} = 1$ ($P = 0.93$), $I^2 = 0\%$; Analysis 4.3) or neurologic adverse events (symptomatic) (test for subgroup differences: $\text{Chi}^2 = 2.32$, $\text{df} = 1$ ($P = 0.13$), $I^2 = 56.9\%$; Analysis 4.4).

We performed a subgroup analysis by sex and stroke severity using disaggregated data. We found no differences by sex for neurologic adverse events (any ICH) (test for subgroup differences: $\text{Chi}^2 = 1.73$, $\text{df} = 1$ ($P = 0.19$), $I^2 = 42.1\%$; Analysis 5.3) or neurologic adverse events (symptomatic) (test for subgroup differences: $\text{Chi}^2 = 1.11$, $\text{df} = 1$ ($P = 0.29$), $I^2 = 10.0\%$; Analysis 5.4), nor for stroke severity (respectively: test for subgroup differences: $\text{Chi}^2 = 0.04$, $\text{df} = 1$ ($P = 0.85$), $I^2 = 0\%$; Analysis 6.3; $\text{Chi}^2 = 0.88$, $\text{df} = 1$ ($P = 0.35$), $I^2 = 0\%$; Analysis 6.4).

Important outcomes

Clinical function (continuous)

For the subgroup analyses, see Analysis 2.5, Analysis 3.5, Analysis 4.5, Analysis 5.5, and Analysis 6.5 in [Supplementary material 7](#).

Investigating treatment effects between different country income levels, we found little to no difference in clinical function (assessed as a continuous variable) (test for subgroup differences: $\text{Chi}^2 = 1.35$, $\text{df} = 1$ ($P = 0.24$), $I^2 = 26.0\%$; Analysis 2.5).

We also did not find a difference in the subgroup analysis according to the intensive blood pressure target in clinical function (assessed as a continuous variable) (test for subgroup differences: $\text{Chi}^2 = 0.43$, $\text{df} = 1$ ($P = 0.51$), $I^2 = 0\%$; Analysis 3.5).

Regarding the subgroup analysis according to the type of procedure performed, we found no difference in the analysis for clinical function (assessed as a continuous variable) (test for subgroup differences: $\text{Chi}^2 = 0.43$, $\text{df} = 1$ ($P = 0.51$), $I^2 = 0\%$; Analysis 4.5).

Lastly, in the subgroup analyses by sex and stroke severity, we also did not find subgroup differences in clinical function (assessed as a continuous variable) (respectively: test for subgroup differences:

$\text{Chi}^2 = 0.31$, $\text{df} = 1$ ($P = 0.58$), $I^2 = 0\%$; Analysis 5.5; $\text{Chi}^2 = 0.14$, $\text{df} = 1$ ($P = 0.71$), $I^2 = 0\%$; Analysis 6.5).

Neurologic status

For the subgroup analyses, see Analysis 2.6, Analysis 2.7, Analysis 3.6, Analysis 3.7, Analysis 4.6, Analysis 4.7, Analysis 5.6, Analysis 5.7, Analysis 6.6, and Analysis 6.7 in [Supplementary material 7](#).

The subgroup analysis according to country income level showed no differences in neurologic status assessed as a continuous outcome (test for subgroup differences: $\text{Chi}^2 = 0.61$, $\text{df} = 1$ ($P = 0.43$), $I^2 = 0\%$) or a dichotomous outcome (test for subgroup differences: $\text{Chi}^2 = 0.18$, $\text{df} = 1$ ($P = 0.67$), $I^2 = 0\%$).

For the subgroup analysis according to the intensive blood pressure target, we also did not find differences in neurologic status assessed as a continuous outcome (test for subgroup differences: $\text{Chi}^2 = 0.04$, $\text{df} = 2$ ($P = 0.98$), $I^2 = 0\%$; Analysis 3.6), nor in the subgroup analysis according to the procedure performed (test for subgroup differences: $\text{Chi}^2 = 3.54$, $\text{df} = 1$ ($P = 0.06$), $I^2 = 71.7\%$; Analysis 4.6).

We did not perform a subgroup analysis of neurologic status (dichotomous), as all included studies reporting this outcome were in the < 140 mmHg group (Analysis 3.7) and the thrombectomy group (Analysis 4.7), respectively.

We performed a subgroup analysis by sex and stroke severity using disaggregated data for these outcomes. We found no differences by sex in neurologic status assessed as a continuous outcome (test for subgroup differences: $\text{Chi}^2 = 0.60$, $\text{df} = 1$ ($P = 0.44$), $I^2 = 0\%$; Analysis 5.6) or a dichotomous outcome (test for subgroup differences: $\text{Chi}^2 = 0.06$, $\text{df} = 1$ ($P = 0.80$), $I^2 = 0\%$; Analysis 5.7), nor by stroke severity (test for subgroup differences: $\text{Chi}^2 = 0.05$, $\text{df} = 1$ ($P = 0.82$), $I^2 = 0\%$; Analysis 6.6; test for subgroup differences: $\text{Chi}^2 = 0.75$, $\text{df} = 1$ ($P = 0.39$), $I^2 = 0\%$; Analysis 6.7, respectively).

All-cause mortality

For the subgroup analyses, see Analysis 2.8, Analysis 3.8, Analysis 3.8, Analysis 4.8, Analysis 5.8, and Analysis 6.8 in [Supplementary material 7](#).

Although all-cause mortality was likely increased in both subgroups, based on country income level, the results differed between subgroups (test for subgroup differences: $\text{Chi}^2 = 8.31$, $\text{df} = 1$ ($P = 0.004$), $I^2 = 88.0\%$), with less favorable outcomes for high-income country studies (high-income countries RR 1.32, 95% CI 1.13 to 1.72; 6 studies, 904 participants; middle-income countries RR 1.10, 95% CI 0.93 to 1.30; 2 studies, 1197 participants; Analysis 2.8).

For the subgroup analysis according to intensive blood pressure target, we found no difference (test for subgroup differences: $\text{Chi}^2 = 1.53$, $\text{df} = 2$ ($P = 0.47$), $I^2 = 0\%$; Analysis 3.8). We also found no subgroup difference for this outcome according to the procedure performed (test for subgroup differences: $\text{Chi}^2 = 0.00$, $\text{df} = 1$ ($P = 0.97$), $I^2 = 0\%$; Analysis 4.8), sex of participants (test for subgroup differences: $\text{Chi}^2 = 0.68$, $\text{df} = 1$ ($P = 0.41$), $I^2 = 0\%$; Analysis 5.8), or stroke severity (test for subgroup differences: $\text{Chi}^2 = 0.67$, $\text{df} = 1$ ($P = 0.41$), $I^2 = 0\%$; Analysis 6.8).

Hospital length of stay

For the subgroup analyses, see Analysis 2.9, Analysis 3.9, Analysis 4.9, Analysis 5.9, and Analysis 6.9 in [Supplementary material 7](#).

The test for subgroup differences indicated no subgroup effect by country income level (test for subgroup differences: $\text{Chi}^2 = 0.15$, $\text{df} = 1$ ($P = 0.70$), $I^2 = 0\%$; Analysis 2.9), procedure performed (test for subgroup differences: $\text{Chi}^2 = 3.62$, $\text{df} = 1$ ($P = 0.06$), $I^2 = 72.4\%$; Analysis 4.9), sex of participants (test for subgroup differences: $\text{Chi}^2 = 1.26$, $\text{df} = 1$ ($P = 0.26$), $I^2 = 20.7\%$; Analysis 5.9), or stroke severity (test for subgroup differences: $\text{Chi}^2 = 1.03$, $\text{df} = 1$ ($P = 0.31$), $I^2 = 3.2\%$; Analysis 6.9). We did find a subgroup difference according to blood pressure target (test for subgroup differences: $\text{Chi}^2 = 6.98$, $\text{df} = 2$ ($P = 0.03$), $I^2 = 71.3\%$; Analysis 3.9), resulting in worse outcomes for the group targeting < 120 mmHg (RR 1.35, 95% CI -4.70 to 7.39; 2 studies, 896 participants), than in those targeting < 140 mmHg (RR -0.80, 95% CI -9.28 to 7.68; 2 studies, 2121 participants) and < 160 mmHg (RR 0, 95% CI -3.71 to 3.71; 1 study, 69 participants).

Other adverse events

For the subgroup analyses, see Analysis 2.10, Analysis 3.10, Analysis 4.10, Analysis 5.10, and Analysis 6.10 in [Supplementary material 7](#).

The test for subgroup differences indicated a subgroup difference by country income level (test for subgroup differences: $\text{Chi}^2 = 4.99$, $\text{df} = 1$ ($P = 0.03$), $I^2 = 80.0\%$; Analysis 2.10), with worse results in the intensive systolic blood pressure group in studies conducted in high-income countries (RR 2.27, 95% CI 1.16 to 4.43; 5 studies, 617 participants) in comparison with studies conducted in middle-income countries (RR 1.19, 95% CI 0.16 to 8.93; 2 studies, 1199 participants).

We did not find a subgroup difference in other adverse events according to the intensive blood pressure target (test for subgroup differences: $\text{Chi}^2 = 0.19$, $\text{df} = 2$ ($P = 0.91$), $I^2 = 0\%$; Analysis 3.10).

However, in the subgroup analysis according to the procedure performed, we found a difference between the two subgroups (test for subgroup differences: $\text{Chi}^2 = 6.93$, $\text{df} = 1$ ($P = 0.008$), $I^2 = 85.6\%$; Analysis 4.10). Although intensive systolic blood pressure management may result in little to no clinically meaningful difference in other adverse events and the evidence was very uncertain, the subgroup analysis (with high imprecision) suggests that systolic blood pressure management may increase other adverse events in the thrombectomy group (RR 1.44, 95% CI 0.90 to 2.30; 6 studies, 1747 participants), but not in the thrombolysis group (RR 0.89, 95% CI 0.67 to 1.18; 2 studies, 2265 participants).

We also conducted a subgroup analysis according to the sex of the included participants (test for subgroup differences: $\text{Chi}^2 = 0.47$, $\text{df} = 1$ ($P = 0.49$), $I^2 = 0\%$; Analysis 5.10).

Finally, we performed a subgroup analysis by stroke severity (test for subgroup differences: $\text{Chi}^2 = 144.00$, $\text{df} = 1$ ($P < 0.00001$), $I^2 = 99.3\%$; Analysis 6.10). We found that participants with a lower baseline NIHSS score had fewer adverse events in the conventional group (RR 3.49, 95% CI 1.74 to 6.99, 2 studies, 62 participants) versus the participants with a higher NIHSS score (RR 0.24, 95% CI 0.02 to 3.74, 2 studies, 37 participants).

3. Conventional versus intensive blood pressure management – sensitivity analyses

We were unable to conduct a sensitivity analysis based on study design, as all included studies were parallel RCTs, nor by excluding unpublished data, since all the included data for our critical outcomes were published.

We conducted a sensitivity analysis for our critical outcomes for the comparison of conventional versus intensive blood pressure management excluding studies at a high risk of bias.

For the clinical function (dichotomous) outcome, the sensitivity analysis, excluding studies at high risk of bias, showed that our results remained robust and precise (RR 0.87, 95% CI 0.78 to 0.99; 7 studies, 4173 participants; Analysis 1.1). We did not conduct a sensitivity analysis for quality of life because none of the included studies were judged to have an overall high risk of bias. Finally, for neurologic adverse events (Analysis 1.3) and neurologic adverse events (symptomatic) (Analysis 1.4), the results also remain robust (RR 0.99, 95% CI 0.84 to 1.15; 6 studies, 3901 participants; RR 1.04, 95% CI 0.76 to 1.42; 7 studies, 4205 participants, respectively).

Equity assessment

Distribution of effects across 'PROGRESS-PLUS' characteristics

We conducted an equity subgroup analysis, as prespecified, according to participants' sex and the socioeconomic status of the countries where the studies were conducted (based on the World Bank classification of income countries).

We created a table showing a summary of the characteristics of participants we should expect to see in the evidence, and the actual participants' characteristics extracted from the included studies ([Supplementary material 9](#)).

None of the studies were conducted in low-income countries. Two of the studies were conducted in middle-income countries (ENCHANTED2/MT; IDENTIFY), six in high-income countries (Avidzba 2024; BP-TARGET; CLEVER; DETECT; OPTIMAL-BP; The BEST-II), and the last included both middle-income and high-income countries (ENCHANTED). Since the ENCHANTED study was conducted in countries with different income statuses and no disaggregated country data were available, we could not include this study in the subgroup analysis.

Regarding participants' sex, all studies included both sexes in varying distributions, with female participants ranging from 37% to 70%. However, not all studies reported data disaggregated by sex to allow subgroup analyses. Only the authors of the studies Avidzba 2024, DETECT, IDENTIFY, and OPTIMAL-BP kindly provided the data needed to perform the analysis. Additionally, the ENCHANTED study reported dichotomous data on clinical function.

For more details about the equity assessment by sex and socioeconomic status, see above (2. Conventional versus intensive blood pressure management — subgroup analyses). More information about subgroups is available in: Analysis 2.1; Analysis 2.10; Analysis 2.2; Analysis 2.3; Analysis 2.4; Analysis 2.5; Analysis 2.6; Analysis 2.7; Analysis 2.8; Analysis 2.9; Analysis 5.1; Analysis 5.10; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 5.6; Analysis 5.7; Analysis 5.8; Analysis 5.9 in [Supplementary material 7](#).

We did not create a summary of findings table based on the equity assessment because we found no differences in subgroup analyses by sex or socio-economic status in the available data for our critical outcomes.

Reporting biases

We did not conduct a reporting bias assessment, as we could not include more than 10 trials.

DISCUSSION

Summary of main results

We included nine RCTs with 4381 participants that compared intensive versus conventional blood pressure management. All the studies had a follow-up period of 90 days, 12 weeks, or three months. Seven studies targeted participants who underwent endovascular thrombectomy, and two targeted participants who underwent systemic thrombolysis as the reperfusion method. Five studies used a target blood pressure of < 140 mmHg as the intervention, two targeted < 120 mmHg, another < 160 mmHg, and the last included two intervention arms with different targets (< 140 mmHg and < 160 mmHg).

The synthesized evidence from our comparison indicates that, compared with conventional blood pressure management, intensive blood pressure management probably results in minimal to no clinically meaningful differences in quality of life, clinical function (assessed continuously with the mRS), neurologic status (evaluated continuously with the NIHSS), or hospital length of stay. It also results in little to no clinically meaningful difference in clinical function (assessed dichotomously as a favorable outcome for participants with an mRS of 0 to 2). It does not increase or decrease the risk of neurologic adverse events (assessed as any ICH). Additionally, it may result in little to no meaningful difference in symptomatic neurologic adverse events (symptomatic ICH). Furthermore, it likely decreases the number of participants with a favorable neurologic status (assessed dichotomously using the NIHSS scale) and likely increases all-cause mortality. Finally, it may have little to no clinically meaningful effect on other adverse events (other than ICH), but the evidence is very uncertain.

Limitations of the evidence included in the review

We identified some uncertainties during this review. All the included studies were conducted in middle- and high-income countries, with none in low-income countries. Almost no patient-reported outcomes were included; only three studies evaluated quality of life. The main reasons for downgrading evidence certainty (GRADE) were the high risk of bias in some included studies and the imprecision in some results. The main risks of bias identified were deviations from the intended interventions, as one study excluded participants based on adherence to the protocol; missing outcome data, as one study had an asymmetrical number of dropouts between groups; and selection of the reported result, due to the presence of some discrepancies between the published protocols or registry and the final published results. There were also concerns about the risk of bias in the measurement of the quality of life outcome, as the participants who served as assessors for this outcome were not blinded. However, the nature of the intervention and comparison is unlikely to have influenced their answers. We also found some imprecision in the results for neurologic adverse events (symptomatic), neurologic status (dichotomous), mortality, and other adverse events. Finally, we were unable to look for publication bias.

Limitations of the review processes

We conducted a comprehensive search across multiple databases and registries and identified all relevant studies, secondary references, and unpublished data. We tracked registered records, authors related to the study, and conference reports. Through these extensive searches and classifications, we believe it is unlikely that we have missed any relevant studies or outcomes. We failed to classify four studies, which remain awaiting classification: one due to the inability to obtain the full text, and the other four because their status is unknown. We tried to clarify the status of these studies, but we received no response from the study authors. We followed standard Cochrane methods for screening, data extraction, and risk of bias assessment using the Cochrane RoB 2 tool.

We intended to conduct subgroup analyses to explore heterogeneity based on predefined criteria, without making decisions based on data inspection, but rather on the evidentiary requirements for decision-making. We contacted the authors to inquire about the possibility of obtaining the disaggregated data for some of our subgroup analyses. Although we received a kind response and the requested information from half of the study authors, the other half did not respond to our email, limiting the available data for the subgroup analyses by sex, stroke severity, and country income level, which probably impacted the results.

Additionally, the inclusion of populations treated with different reperfusion modalities has contributed to heterogeneity in the treatment effects for some outcomes. To address this, and in accordance with our protocol, we conducted prespecified subgroup analyses by the primary procedure performed in the included studies. Finally, other factors, such as reperfusion timing, vascular territory, infarct size, baseline blood pressure, blood pressure variability, and antihypertensive strategy, which might contribute to heterogeneity, were not prespecified as subgroup analyses in our protocol.

Agreements and disagreements with other studies or reviews

This review provides new insights by focusing specifically on blood pressure targets after reperfusion in acute ischemic stroke. Earlier Cochrane reviews have primarily evaluated the benefits and risks of different reperfusion strategies and modalities, variations in thrombolytic regimens, and diverse endovascular approaches across broader stroke populations [5, 5, 8, 9, 178, 179, 180]. In contrast, we restricted inclusion to patients who received intravenous thrombolysis and/or endovascular thrombectomy and directly compared intensive versus conventional post-reperfusion systolic blood pressure management.

Current international guidelines recommend relatively conservative blood pressure thresholds after reperfusion, mainly based on observational data, expert consensus, and pre-thrombectomy trials. The American Heart Association/American Stroke Association (AHA/ASA) guidelines advise lowering blood pressure to < 185/110 mmHg before intravenous thrombolysis and maintaining blood pressure < 180/105 mmHg for 24 hours after alteplase and/or thrombectomy [15]. The European Stroke Organisation (ESO) guidelines similarly support targets below 180 mmHg while acknowledging the uncertainty and limited quality of evidence [16]. Consistent with these positions, the updated

Korean guideline on blood pressure management in acute stroke synthesized some of the recent post-thrombectomy trials and concluded that systolic blood pressure should be maintained at or below 180 mmHg during the first 24 hours after successful endovascular reperfusion therapy, and that more aggressive blood pressure lowering is not recommended because it offers no clinical benefit and may be harmful [181]. Our findings are broadly consistent with this cautious stance, that maintaining systolic blood pressure around or below 180 mmHg after reperfusion appears reasonable, whereas more aggressive lowering is unlikely to improve functional outcome and may be harmful, in line with recent narrative reviews [182, 183].

Observational studies have often been interpreted as supporting a “lower is better” paradigm, as higher mean or peak blood pressure after thrombectomy has been associated with larger infarcts, more symptomatic intracranial hemorrhage, and worse functional outcomes [17, 19, 184]. However, these analyses usually treat blood pressure as an attained level and rarely distinguish spontaneous blood pressure declines from pharmacologically induced reductions. Patients whose blood pressure falls spontaneously after successful reperfusion may have smaller infarcts, better collaterals and more stable hemodynamics, whereas those requiring active blood pressure lowering are typically more unstable or have more severe strokes. Blood pressure thus acts both as a prognostic marker and a treatment target, and confounding by indication is likely even after statistical adjustment. Our review, based exclusively on randomized controlled trials, suggests that intensive post-reperfusion blood pressure lowering does not convert the observational association between high blood pressure and poor outcomes into clinical benefit and may instead increase mortality and unfavorable neurological status. This supports the hypothesis that much of the apparent benefit of “lower blood pressure” in observational cohorts reflects the benign course of patients with spontaneous blood pressure decline rather than a causal effect of active blood pressure reduction.

Our findings also align with and extend previous systematic and narrative reviews of blood pressure management in acute ischemic stroke, particularly in the endovascular era. Earlier reviews dominated by observational data concluded that higher post-procedural blood pressure and greater blood pressure variability are associated with worse outcomes, but highlighted heterogeneity in blood pressure targets and the lack of robust randomized evidence [12, 184, 185, 186]. More recent quantitative syntheses that incorporate post-thrombectomy trials report that intensive blood pressure lowering may reduce some hemorrhagic events but does not improve functional outcome or mortality, and may be associated with lower functional independence when very low targets are pursued [182, 183, 187]. Overall, current trial-focused evidence suggests that high blood pressure and blood pressure variability are markers of poor prognosis. Still, aggressive early blood pressure lowering, particularly when rapid and pharmacologically driven, has not improved outcomes and may be harmful in reperfused patients. Our review supports this interpretation and underlines the need to distinguish between spontaneous blood pressure declines, which may signal successful reperfusion and preserved autoregulation, and active blood pressure-lowering interventions, which may compromise perfusion in vulnerable tissue.

AUTHORS' CONCLUSIONS

Implications for practice

Compared with conventional blood pressure management, intensive blood pressure management likely results in little to no clinically meaningful differences in quality of life, clinical function (assessed continuously with the modified Rankin Scale), neurologic status (evaluated continuously with the National Institutes of Health Stroke Scale), and hospital length of stay. It also results in little to no clinically meaningful difference in clinical function (dichotomous) and does not increase or reduce neurologic adverse events. Additionally, it may result in little to no clinically meaningful difference in symptomatic neurologic adverse events, and it may have little to no clinically meaningful effect on other adverse events (other than intracranial hemorrhage). Lastly, it probably reduces the number of participants with a favorable neurologic status (assessed as a dichotomous outcome) and probably increases all-cause mortality.

The certainty of the evidence for most outcomes was moderate, except for clinical function as a dichotomous outcome and neurologic adverse events, which we assessed as high; neurologic adverse events (symptomatic), which we rated as low; and other adverse events, which we rated as very low.

Equity-related implications for practice

We conducted prespecified equity analyses by sex and country income level. Our analyses suggest that, overall, women were reasonably represented in the included trials, with around 47% of participants. In the limited sex-disaggregated data available, we did not detect important differences between women and men in the effects of intensive versus conventional systolic blood pressure targets on functional outcome, quality of life, mortality, or neurological adverse events. On current evidence, there is no justification for sex-specific blood pressure targets after reperfusion.

By contrast, the geographic distribution of evidence is strongly skewed towards high-income settings, with no trials conducted in low-income countries and only a minority in middle-income countries. Implementing intensive post-reperfusion blood pressure protocols in health systems with limited access to reperfusion therapies, continuous monitoring, and intravenous antihypertensives may not be feasible. Clinicians and policymakers in such settings should therefore adapt our findings to local capacity and avoid importing protocols that are resource-intensive but of uncertain benefit in their context.

Implications for research

Our findings suggest that intensive blood pressure targets after successful reperfusion, often set below 120 mmHg to 140 mmHg, do not improve functional outcomes and may increase harm compared with conventional targets below 180 mmHg. Future trials, if ethical, should focus on more moderate blood pressure strategies and carefully assess early neurological deterioration, hemorrhagic transformation, infarct growth, and long-term outcomes.

New studies should pre-register protocols, clearly describe randomization methods, and standardize outcome definitions to enable effective data pooling for meta-analyses. It is also

crucial to identify which patients may benefit from different blood pressure strategies. Future research should include subgroup analyses based on age, baseline blood pressure, co-interventions, prior hypertension, stroke severity, and risk of hemorrhagic transformation, while reporting sex-disaggregated and other equity-relevant data. Additionally, incorporating imaging and physiological markers could lead to more individualized blood pressure targets.

Finally, future trials and well-designed observational studies should recruit participants from low- and middle-income countries and report contextual factors such as access to reperfusion and drug availability. Using modern analytical methods will help address confounding factors and ensure that future blood pressure management guidelines for reperfused ischemic stroke are methodologically robust and globally applicable.

Equity-related implications for research

Future trials of blood pressure management after reperfused ischemic stroke should be designed and reported with equity in mind. At a minimum, they should routinely collect and report sex-disaggregated baseline and outcome data and be sufficiently powered to detect meaningful sex differences. Trials should also expand recruitment beyond high-income tertiary centers to include middle- and, where feasible, low-income countries, and explicitly describe context-relevant factors such as access to reperfusion, monitoring capacity, and drug availability. Incorporating equity frameworks (for example, PROGRESS-PLUS) into trial design and reporting would improve assessment of who is represented in the evidence and help ensure that future guidelines on post-reperfusion blood pressure management do not unintentionally widen global inequities in stroke care.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD016085.pub2](https://doi.org/10.1002/14651858.CD016085.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Risk of bias

Supplementary material 7 Analyses

Supplementary material 8 Data package

Supplementary material 9 Summary of the characteristics of participants we should expect to see in the evidence and the actual participants' characteristics extracted from the included studies

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Toby Lasserson, Cochrane;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Anupa Shah, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments, and supported the editorial team): Andrew Savage, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Jenny Bellorini, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Craig Anderson, The George Institute for Global Health, UNSW, Sydney, Australia (clinical/content review); Tejaswini Arunachala Murthy (patient and public review); Clare Miles, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Cochrane Central Editorial Information Specialist (search review).

One additional peer reviewer provided a clinical peer review but chose not to be publicly acknowledged.

Contributions of authors

All review authors read and approved the final draft of the review.

LBV conceived and conceptualized the review.

CE, LBV: developed the search strategy.

LBV, SD, CE, MAB, DI, JPR, EQ, NM, and LIG participated in the methodology, writing, review, and editing.

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Registration and protocol

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The protocol was approved and published on 4 March 2025. Protocol available via doi.org/10.1002/14651858.CD016085.

Data, code and other materials

As part of the published Cochrane review, the following will be made available for download for users of the Cochrane Library: full search strategies for each database ([Supplementary material 1](#)); full citations of each unique report for all studies included ([Supplementary material 2](#)), ongoing or awaiting classification ([Supplementary material 4](#); [Supplementary material 5](#)), or excluded at the full-text screen ([Supplementary material 3](#)), in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows ([Supplementary material 6](#); [Supplementary material 7](#); [Supplementary material 8](#)). Appropriate permissions will be obtained for such use. Analyses and data management will be conducted using Cochrane's authoring tool, Review Manager Web, using built-in computation methods.

Disclosure of artificial intelligence use

Grammarly (Grammarly Inc., San Francisco, CA, USA) was used for language editing, including grammar, spelling, and style. No artificial intelligence tools were employed to generate, analyze, or interpret data, or to draft the scientific content of this review. All suggestions from Grammarly were carefully reviewed and approved by the authors, who take full responsibility for the manuscript's content.

REFERENCES

1. Wu P, Zeng F, Li YX, Yu BL, Qiu LH, Qin W, et al. Changes of resting cerebral activities in subacute ischemic stroke patients. *Neural Regeneration Research* 2015;**10**(5):760-5.
2. Yoon CW, Bushnell CD. Stroke in women: a review focused on epidemiology, risk factors, and outcomes. *Journal of Stroke* 2023;**25**(1):2-15.
3. Li XY, Kong XM, Yang CH, Cheng ZF, Lv JJ, Guo H, et al. Global, regional, and national burden of ischemic stroke, 1990-2021: an analysis of data from the Global Burden of Disease Study 2021. *EClinicalMedicine* 2024;**75**:102758.
4. GBD 2021 Stroke Risk Factor Collaborators. Global, regional, and national burden of stroke and its risk factors, 1990-2021: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet Neurology* 2024;**23**(10):973-1003.
5. Lindekleiv H, Berge E, Bruins Slot KMH, Wardlaw JM. Percutaneous vascular interventions versus intravenous thrombolytic treatment for acute ischaemic stroke. *Cochrane Database of Systematic Reviews* 2018, Issue 10. Art. No: CD009292. [DOI: [10.1002/14651858.CD009292.pub2](https://doi.org/10.1002/14651858.CD009292.pub2)]
6. Tsao CW, Aday AW, Almarzooq ZI, Alonso A, Beaton AZ, Bittencourt MS, et al. Heart disease and stroke statistics - 2022 update: a report from the American Heart Association. *Circulation* 2022;**145**(8):e153-639.
7. Ogawa A, Mori E, Minematsu K, Taki W, Takahashi A, Nemoto S, et al; MELT Japan Study Group. Randomized trial of intraarterial infusion of urokinase within 6 hours of middle cerebral artery stroke: the middle cerebral artery embolism local fibrinolytic intervention trial (MELT) Japan. *Stroke* 2007;**38**(10):2633-9.
8. Roaldsen MB, Jusufovic M, Berge E, Lindekleiv H. Endovascular thrombectomy and intra-arterial interventions for acute ischaemic stroke. *Cochrane Database of Systematic Reviews* 2021, Issue 6. Art. No: CD007574. [DOI: [10.1002/14651858.CD007574.pub3](https://doi.org/10.1002/14651858.CD007574.pub3)]
9. Wardlaw JM, Murray V, Berge E, Del Zoppo GJ. Thrombolysis for acute ischaemic stroke. *Cochrane Database of Systematic Reviews* 2014, Issue 7. Art. No: CD000213. [DOI: [10.1002/14651858.CD000213.pub3](https://doi.org/10.1002/14651858.CD000213.pub3)]
10. Mistry EA, Hart KW, Davis LT, Gao Y, Prestigiacomo CJ, Mittal S, et al. Blood pressure management after endovascular therapy for acute ischemic stroke: the BEST-II randomized clinical trial. *JAMA* 2023;**330**(9):821-31.
11. Mazighi M, Richard S, Lapergue B, Sibon I, Gory B, Berge J, et al. Safety and efficacy of intensive blood pressure lowering after successful endovascular therapy in acute ischaemic stroke (BP-TARGET): a multicentre, open-label, randomised controlled trial. *Lancet Neurology* 2021;**20**(4):265-74.
12. De Georgia M, Bowen T, Duncan KR, Chebl AB. Blood pressure management in ischemic stroke patients undergoing mechanical thrombectomy. *Neurological Research and Practice* 2023;**5**(1):12.
13. Malhotra K, Ahmed N, Filippatou A, Katsanos AH, Goyal N, Tsioufis K, et al. Association of elevated blood pressure levels with outcomes in acute ischemic stroke patients treated with intravenous thrombolysis: a systematic review and meta-analysis. *Journal of Stroke* 2019;**21**(1):78-90.
14. Nam HS, Kim YD, Heo J, Lee H, Jung JW, Choi JK, et al; OPTIMAL-BP Trial Investigators. Intensive vs conventional blood pressure lowering after endovascular thrombectomy in acute ischemic stroke: the OPTIMAL-BP randomized clinical trial. *JAMA* 2023;**330**(9):832-42.
15. Powers WJ, Rabinstein AA, Ackerson T, Adeoye OM, Bambakidis NC, Becker K, et al. Guidelines for the early management of patients with acute ischemic stroke: 2019 update to the 2018 guidelines for the early management of acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* 2019;**50**(12):e344-418.
16. Sandset EC, Anderson CS, Bath PM, Christensen H, Fischer U, Gasecki D, et al. European Stroke Organisation (ESO) guidelines on blood pressure management in acute ischaemic stroke and intracerebral haemorrhage. *European Stroke Journal* 2021;**6**(2):XLVIII-LXXXIX.
17. Mistry EA, Mayer SA, Khatri P. Blood pressure management after mechanical thrombectomy for acute ischemic stroke: a survey of the StrokeNet sites. *Journal of Stroke* 2018;**27**(9):2474-8.
18. Anadani M, Lapergue B, Blanc R, Kyheng M, Labreuche J, Machaa MB, et al. Admission blood pressure and outcome of endovascular therapy: secondary analysis of ASTER trial. *Journal of Stroke and Cerebrovascular Diseases* 2020;**29**(12):105347.
19. Mistry EA, Sucharew H, Mistry AM, Mehta T, Arora N, Starosciak AK, et al. Blood pressure after endovascular therapy for ischemic stroke (BEST): a multicenter prospective cohort study. *Stroke* 2019;**50**(12):3449-55.
20. Yang P, Song L, Zhang Y, Zhang X, Chen X, Li Y, et al. Intensive blood pressure control after endovascular thrombectomy for acute ischaemic stroke (ENCHANTED2/MT): a multicentre, open-label, blinded-endpoint, randomised controlled trial. *Lancet* 2022;**400**(10363):1585-96.
21. Zhou Z, Xia C, Carcel C, Yoshimura S, Wang X, Delcourt C, et al. Intensive versus guideline-recommended blood pressure reduction in acute lacunar stroke with intravenous thrombolysis therapy: the ENCHANTED trial. *European Journal of Neurology* 2021;**28**(3):783-93.
22. Cumpston M, Lasserson T, Fleming E, Page MJ. Chapter III: Reporting the review. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.

23. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ (Clinical Research Edition)* 2020;**372**(71):n71. [DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71)]
24. Schünemann H, Brożek J, Guyatt G, Oxman A, editor(s). Handbook for grading the quality of evidence and the strength of recommendations using the GRADE approach (updated October 2013). GRADE Working Group, 2013. Available from <https://gdt.gradepro.org/app/handbook/handbook.html>.
25. Higgins JP, Eldridge S, Li T. Chapter 23: Including variants on randomized trials. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
26. Reeves BC, Deeks JJ, Higgins JP, Shea B, Tugwell P, Wells GA. Chapter 24: Including non-randomized studies on intervention effects. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
27. Wilson A, Bath PM, Berge E, Cadilhac DA, Cuche M, Ford GA, et al. Understanding the relationship between costs and the modified Rankin Scale: a systematic review, multidisciplinary consensus and recommendations for future studies. *European Stroke Journal* 2017;**2**(1):3-12.
28. Mishra B, Sudheer P, Agarwal A, Srivastava MV, Nilima, Vishnu VY. Minimal clinically important difference (MCID) in patient-reported outcome measures for neurological conditions: review of concept and methods. *Annals of Indian Academy of Neurology* 2023;**26**(4):334-43.
29. Hacke W, Kaste M, Bluhmki E, Brozman M, Dávalos A, Guidetti D, et al. Thrombolysis with alteplase 3 to 4.5 hours after acute ischemic stroke. *New England Journal of Medicine* 2008;**359**(13):1317-29.
30. Lefebvre C, Glanville J, Briscoe S, Featherstone R, Littlewood A, Metzendorf M-I, et al. Chapter 4: Searching for and selecting studies [last updated March 2025]. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5.1. Cochrane, 2025. Available from [cochrane.org/handbook](https://www.cochrane.org/handbook).
31. Covidence. Version accessed 18 February 2024. Melbourne, Australia: Veritas Health Innovation, 2021. Available at <https://www.covidence.org>.
32. Review Manager (RevMan). Version 7.5.0. The Cochrane Collaboration, 2024. Available at <https://revman.cochrane.org>.
33. Hoffmann TC, Glasziou PP, Boutron I, Milne R, Perera R, Moher D, et al. Die TIDieR Checkliste und Anleitung - ein Instrument für eine verbesserte Interventionsbeschreibung und Replikation [Better reporting of interventions: Template for Intervention Description and Replication (TIDieR) checklist and guide]. *Das Gesundheitswesen* 2016;**78**(3):175-88.
34. Hoffmann TC, Glasziou PP, Boutron I, Milne R, Perera R, Moher D, et al. Better reporting of interventions: template for intervention description and replication (TIDieR) checklist and guide. *BMJ (Clinical Research Edition)* 2014;**348**:1687.
35. Higgins JP, Sterne JA, Savović J, Page MJ, Hróbjartsson A, Boutron I. A revised tool for assessing risk of bias in randomized trials. *Cochrane Database of Systematic Reviews* 2016, Issue 10. Art. No: CD201601. [DOI: [10.1002/14651858.CD201601](https://doi.org/10.1002/14651858.CD201601)]
36. Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized trial. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
37. Sterne JA, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ (Clinical Research Edition)* 2019;**366**:14898.
38. Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
39. Higgins JP, Li T, Deeks JJ. Chapter 6: Choosing effect measures and computing estimates of effect. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5. Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
40. Kirkham JJ, Dwan KM, Altman DG, Gamble C, Dodd S, Smyth R, et al. The impact of outcome reporting bias in randomised controlled trials on a cohort of systematic reviews. *BMJ (Clinical Research Edition)* 2010;**340**:c365.
41. Sterne JA, Sutton AJ, Ioannidis JP, Terrin N, Jones DR, Lau J, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ (Clinical Research Edition)* 2011;**343**:d4002.
42. Page MJ, Higgins JP, Sterne JA. Chapter 13: Assessing risk of bias due to missing results in a synthesis. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
43. Wood L, Egger M, Gluud LL, Schulz KF, Jüni P, Altman DG, et al. Empirical evidence of bias in treatment effect estimates in controlled trials with different interventions and outcomes: meta-epidemiological study. *BMJ (Clinical Research Edition)* 2008;**336**(7644):601-5.
44. Deeks JJ, Higgins JP, Altman DG. Chapter 10: Analysing data and undertaking meta-analyses. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.

45. McKenzie JE, Brennan SE. Chapter 12: Synthesizing and presenting findings using other methods. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
46. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ (Clinical Research Edition)* 2003;**327**(414):557-60.
47. Altman DG, Bland JM. Interaction revisited: the difference between two estimates. *BMJ (Clinical Research Edition)* 2003;**326**(7382):219.
48. Schünemann HJ, Higgins JP, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
49. GRADEpro GDT. Version 2024. Hamilton (ON): McMaster University (developed by Evidence Prime), accessed 18 February 2024. Available at <https://www.gradepr.org>.
50. Santesso N, Glenton C, Dahm P, Garner P, Akl EA, Alper B, et al. GRADE guidelines 26: informative statements to communicate the findings of systematic reviews of interventions. *Journal of Clinical Epidemiology* 2020;**119**:126-35.
51. Welch V, Petkovic J, Jull J, Hartling L, Klassen T, Kristjansson E, et al. Chapter 16: Equity and specific populations. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.
52. Avidzba AR, Saskin VA, Nikonov AM, Ayyaz H, Kirov MYu. Arterial pressure during first day after intravenous thrombolysis and acute ischemic stroke outcomes: single-center randomized controlled study [Артериальное давление в первые сутки после внутривенного тромболитического и исходы ишемического инсульта: одноцентровое рандомизированное контролируемое исследование]. *Messenger of Anesthesiology and Resuscitation* 2024;**21**(6):42-50. [DOI: [10.24884/2078-5658-2024-21-6-42-50](https://doi.org/10.24884/2078-5658-2024-21-6-42-50)]
53. NCT05517109. Hemodynamic in postreperfusion period and functional recovery in acute ischemic stroke patients. <https://clinicaltrials.gov/study/NCT05517109> 2022.
54. Avidzba AR, Saskin VA, Nikonov AM, Hussain A, Kirov MY. Linking cerebral oximetry to outcomes of reperfusion therapy in ischemic stroke: a post-hoc analysis of a randomized controlled trial [Взаимосвязь показателей церебральной оксиметрии с исходами реперфузионной терапии при ишемическом инсульте: post hoc анализ рандомизированного контролируемого исследования]. *General Reanimatology* 2024;**20**(6):22-8. [DOI: [10.15360/1813-9779-2024-6-2518](https://doi.org/10.15360/1813-9779-2024-6-2518)]
55. Mazighi M, Richard S, Lapergue B, Sibon I, Gory B, Berge J, et al. Safety and efficacy of intensive blood pressure lowering after successful endovascular therapy in acute ischaemic stroke (BP-TARGET): a multicentre, open-label, randomised controlled trial. *Lancet. Neurology* 2021;**20**(4):265-74. [DOI: [10.1016/S1474-4422\(20\)30483-X](https://doi.org/10.1016/S1474-4422(20)30483-X)]
56. Anadani M, Maier B, Escalard S, Labreuche J, de Havenon A, Sabben C, et al. Safety and efficacy of intensive systolic blood pressure lowering after successful endovascular therapy: a post hoc analysis of the BP TARGET trial. *Journal of Neurointerventional Surgery* 2023;**15**(e1):e142-7. [DOI: [10.1136/jnis-2022-019130](https://doi.org/10.1136/jnis-2022-019130)]
57. Anadani M, Maier B, Escalard S, Labreuche J, de Havenon A, Sabben C, et al. Magnitude of blood pressure change after endovascular therapy and outcomes: insight from the BP-TARGET trial. *Stroke* 2022;**53**(3):719-27. [DOI: [10.1161/STROKEAHA.121.036701](https://doi.org/10.1161/STROKEAHA.121.036701)]
58. Maier B, Gory B, Lapergue B, Sibon I, Richard S, Kyheng M, et al. Effect of baseline antihypertensive treatments on stroke severity and outcomes in the BP TARGET trial. *Stroke* 2022;**53**(6):1837-46. [DOI: [10.1161/STROKEAHA.121.037548](https://doi.org/10.1161/STROKEAHA.121.037548)]
59. Maier B, Brauner R, Escalard S, Gory B, Lapergue B, Sibon I, et al. Association of contrast enhancement after reperfusion with outcomes according to blood pressure lowering in patients with acute ischemic stroke. *Neurology* 2022;**99**(21):e2385-94. [DOI: [10.1212/WNL.000000000000201173](https://doi.org/10.1212/WNL.000000000000201173)]
60. Maier B, Gory B, Lapergue B, Sibon I, Escalard S, Kyheng M, et al. Effect of blood pressure variability in the randomized controlled BP TARGET trial. *European Journal of Neurology* 2022;**29**(3):771-81. [DOI: [10.1111/ene.15194](https://doi.org/10.1111/ene.15194)]
61. Mazighi M, Desilles J-P, Escalard S, Richard S, Gory B, Lapergue B, et al. Blood pressure target in acute stroke to reduce hemorrhage after endovascular therapy: the randomized BP target study. *International Journal of Stroke* 2020;**15**(1 Suppl):9. [DOI: [10.1177/1747493020963387](https://doi.org/10.1177/1747493020963387)]
62. Mazighi M, Labreuche J, Richard S, Gory B, Lapergue B, Sibon I, et al. Blood pressure target in acute stroke to reduce hemorrhage after endovascular therapy: the randomized BP TARGET study protocol. *Frontiers in Neurology* 2020;**11**(101546899):480. [DOI: [10.3389/fneur.2020.00480](https://doi.org/10.3389/fneur.2020.00480)]
63. NCT03160677. Blood Pressure Target in Acute Stroke to Reduce hemorrhage After Endovascular Therapy (BP-TARGET). <http://clinicaltrials.gov/study/NCT03160677> 2017. [HTTP://CLINICALTRIALS.GOV/STUDY/: NCT03160677]
64. Jumaa MA, Gharaibeh K, Burgess RE, Rao R, Oliver MJ, et al. Clevidipine infusion for blood pressure management after successful revascularisation in acute ischaemic stroke: the CLEVER study. *European Stroke Journal* 2026;**11**(2):aakag005. [DOI: [10.1093/esj/aakag005](https://doi.org/10.1093/esj/aakag005)]
65. NCT05175547. Clevidipine infusion for blood pressure management after successful revascularization in acute ischemic stroke. <https://clinicaltrials.gov/study/NCT05175547> 2021.

66. Katsanos AH, Catanese L, Sahlas DJ, Srivastava A, Veroniki AA, Perera K, et al. Blood pressure management following endovascular stroke treatment: a feasibility trial and meta-analysis of outcomes. *Stroke: Vascular and Interventional Neurology* 2024;**4**(3):e001287.
67. NCT04484350. Blood pressure management in stroke following endovascular treatment (DETECT). <https://clinicaltrials.gov/study/NCT04484350> 2020. [HTTPS://CLINICALTRIALS.GOV/STUDY/: NCT04484350]
68. Katsanos A, Catanese L, Sahlas DJ, Srivastava A, Veroniki AA, Perera K, et al. Blood pressure management in stroke following endovascular treatment (DETECT): a feasibility trial and metaanalysis of outcomes. *European Stroke Journal* 2023;**8**(2 Suppl):698. [DOI: [10.1177/23969873231174267](https://doi.org/10.1177/23969873231174267)]
69. Anderson CS, Huang Y, Lindley RI, Chen X, Arima H, Chen G, et al. Intensive blood pressure reduction with intravenous thrombolysis therapy for acute ischaemic stroke (ENCHANTED): an international, randomised, open-label, blinded-endpoint, phase 3 trial. *Lancet* 2019;**393**(10174):877-88. [DOI: [10.1016/S0140-6736\(19\)30038-8](https://doi.org/10.1016/S0140-6736(19)30038-8)]
70. Chalmers J, Anderson C. Efficacy of low-dose vs standard-dose rtPA between Asian and non-Asian with particular emphasis on stroke subtype and risk of intracerebral hemorrhage. *European Stroke Journal* 2016;**1**(1 Suppl):708. [DOI: [10.1177/2396987316642910](https://doi.org/10.1177/2396987316642910)]
71. Delcourt C, Wang X, Zhou Z, Wardlaw J, Mair G, Robinson T, et al. Baseline brain imaging signs and outcomes in the ENCHANTED trial. *European Stroke Journal* 2019;**4**(Supplement 1):418. [DOI: [10.1177/2396987319845581](https://doi.org/10.1177/2396987319845581)]
72. Ren X, Song L, Anderson C, Zhao Y. Influence of prior diabetes mellitus on outcome for thrombolysis-treated acute ischemic stroke: a post hoc analysis of ENCHANTED trial. *European Stroke Journal* 2023;**8**(2 Supplement):174-5. [DOI: [10.1177/23969873231169660](https://doi.org/10.1177/23969873231169660)]
73. Anderson C, Robinson T, Lindley R, Lavados P, Huang Y, Sharma V, et al. ENCHANTED main results - Trial of low-dose versus standard-dose intravenous tissue plasminogen activator in 3310 patients with acute ischemic stroke. *European Stroke Journal* 2016;**1**(1 Suppl):707. [DOI: [10.1177/2396987316642910](https://doi.org/10.1177/2396987316642910)]
74. Anderson CS, Robinson T, Lindley RI, Arima H, Lavados PM, Lee TH, et al. Low-dose versus standard-dose intravenous alteplase in acute ischemic stroke. *New England Journal of Medicine* 2016;**374**(24):2313-23. [DOI: [10.1056/NEJMoa1515510](https://doi.org/10.1056/NEJMoa1515510)]
75. Anderson CS, Woodward M, Arima H, Chen X, Lindley RI, Wang X, et al. Statistical analysis plan for evaluating low- vs. standard-dose alteplase in the enhanced control of hypertension and thrombolysis stroke study (ENCHANTED). *International Journal of Stroke* 2015;**10**(8):1313-5. [DOI: [10.1111/ijls.12602](https://doi.org/10.1111/ijls.12602)]
76. Anderson CS, Woodward M, Arima H, Chen X, Lindley RI, Wang X, et al. Statistical analysis plan for evaluating different intensities of blood pressure control in the enhanced control of hypertension and thrombolysis stroke study. *International Journal of Stroke* 2019;**14**(5):555-8. [DOI: [10.1177/1747493018806170](https://doi.org/10.1177/1747493018806170)]
77. Beishon L, Wang X, Batterham A, Anderson CS, Robinson TG. Bodyweight and rates of ICH according to alteplase dose in patients on prior antiplatelet therapy: ENCHANTED trial results. *European Stroke Journal* 2019;**4**(Suppl 1):243. [DOI: [10.1177/2396987319845581](https://doi.org/10.1177/2396987319845581)]
78. Carcel C, Else Charlotte S, Wang X, Sato S, Delcourt C, Yoshimura S, et al. Women and stroke sex differences in response to thrombolysis in acute ischemic stroke: ENCHANTED results. *European Stroke Journal* 2017;**2**(1 Suppl):65-6. [DOI: [10.1177/2396987317705236](https://doi.org/10.1177/2396987317705236)]
79. Carcel C, Ouyang M, Wang X, Zhou Z, Delcourt C, Lindley R, et al. Examination of sex differences after acute ischaemic stroke using cerebral imaging. *International Journal of Stroke* 2022;**17**(3 Suppl):38. [DOI: [10.1177/17474930221125973](https://doi.org/10.1177/17474930221125973)]
80. Carr SJ, Wang X, Olavarria VV, Lavados PM, Rodriguez JA, Kim JS, et al. Influence of renal impairment on outcome for thrombolysis-treated acute ischemic stroke: ENCHANTED (enhanced control of hypertension and thrombolysis stroke study) post hoc analysis. *Stroke* 2017;**48**(9):2605-9. [DOI: [10.1161/STROKEAHA.117.017808](https://doi.org/10.1161/STROKEAHA.117.017808)]
81. Chalmers J, Wang X, Moullaali T, Minhas J, Anderson C, Robinson T. Associations of early blood pressure control and outcome in thrombolysis-eligible acute ischaemic stroke in the enhanced control of hypertension and thrombolysis stroke study. *Journal of Hypertension* 2019;**37**(Suppl 1):e91.
82. Chen C, Ouyang M, Ong S, Zhang L, Zhang G, Delcourt C, et al. Effects of intensive blood pressure lowering on cerebral ischaemia: insights from the ENCHANTED trial. *European Stroke Journal* 2022;**7**(1 Suppl):578. [DOI: [10.1177/23969873221094907](https://doi.org/10.1177/23969873221094907)]
83. Chen C, Wang X, Chen X, Ouyang M, Sun L, Arima H, et al. Asians versus non-Asian, differences in characteristic, management and outcomes for tpa-treated acute ischemic stroke patients in the ENCHANTED trial. *International Journal of Stroke* 2020;**15**(1 Suppl):53. [DOI: [10.1177/1747493020963387](https://doi.org/10.1177/1747493020963387)]
84. Chen C, Ouyang M, Ong S, Zhang L, Zhang G, Delcourt C, et al. Effects of intensive blood pressure lowering on cerebral ischaemia in thrombolysed patients: insights from the ENCHANTED trial. *EClinicalMedicine* 2023;**57**(101733727):101849. [DOI: [10.1016/j.eclinm.2023.101849](https://doi.org/10.1016/j.eclinm.2023.101849)]
85. Chen C, Ren X, Zhao Y, Ouyang M, Li Mbiostat Q, Wang X, et al. Intensive blood pressure lowering and renal function in ischemic stroke patients: secondary analysis of the ENCHANTED trial. *Cerebrovascular Diseases (Basel, Switzerland)* 2024;**53**(6):703-10. [DOI: [10.1159/000536275](https://doi.org/10.1159/000536275)]
86. Chen C, Song L, Yang J, Lindley R, Robinson T, Arima H, et al. Regional differences in early BP management after acute ischemic stroke in the ENCHANTED international randomized controlled trials. *Frontiers in Neurology* 2021;**12**(101546899):687862. [DOI: [10.3389/fneur.2021.687862](https://doi.org/10.3389/fneur.2021.687862)]

87. Chen C, Wang X, Chen X, Ouyang M, Sun L, Arima H, et al. Disparities between Asian and non-Asian thrombolized acute ischemic stroke patients in the enhanced control of hypertension and thrombolysis stroke trial. *Cerebrovascular Diseases (Basel, Switzerland)* 2021;**50**(5):560-6. [DOI: [10.1159/000516487](https://doi.org/10.1159/000516487)]
88. Chen G, Wang X, Robinson T, Lindley R, Zhou S, Ping L, et al. Low-dose versus standard-dose alteplase by prior stroke and diabetes mellitus in acute ischemic stroke: the ENCHANTED trial. *European Stroke Journal* 2017;**2**(1 Supplement 1):9. [DOI: [10.1177/2396987317705236](https://doi.org/10.1177/2396987317705236)]
89. Chen X, Anderson C, Robinson T, Lavados P, Fuentes S, Lindley R, et al. The enhanced control of hypertension and thrombolysis stroke study (ENCHANTED): complexity of set-up large international clinical trial. *Cerebrovascular Diseases (Basel, Switzerland)* 2013;**36**(Suppl 1):20. [DOI: [10.1159/000353795](https://doi.org/10.1159/000353795)]
90. Chen X, Anderson CS, Robinson T, Lavados P, Fuentes S, Lindley R, et al. The enhanced control of hypertension and thrombolysis stroke study (ENCHANTED): Complexity of set-up large international clinical trial. *International Journal of Stroke* 2013;**8**(Suppl 1):36-7. [DOI: [10.1111/ij.s.12143](https://doi.org/10.1111/ij.s.12143)]
91. Chen X, Delcourt C, Sun L, Zhou Z, Yoshimura S, You S, et al. Brain imaging signs and health-related quality of life after acute ischemic stroke: analysis of ENCHANTED alteplase dose arm. *Cerebrovascular Diseases (Basel, Switzerland)* 2020;**49**(4):427-36. [DOI: [10.1159/000509226](https://doi.org/10.1159/000509226)]
92. Chen X, Wang X, Delcourt C, Li J, Arima H, Hackett ML, et al. Ethnicity and other determinants of quality of functional outcome in acute ischemic stroke: the ENCHANTED trial. *Stroke* 2020;**51**(2):588-93. [DOI: [10.1161/STROKEAHA.119.027639](https://doi.org/10.1161/STROKEAHA.119.027639)]
93. Diener HC, Nitschmann S. Optimal blood pressure control in patients with acute ischaemic stroke: ENCHANTED trial. *Der Internist* 2019;**60**(10):1118-20. [DOI: [10.1007/s00108-019-00659-1](https://doi.org/10.1007/s00108-019-00659-1)]
94. Fuentes S, Huang Y, Wang J, Sharma V, Nguyen HT, Pandian J, et al. The enhanced control of hypertension and thrombolysis stroke study (ENCHANTED): Part A-rationale for a trial of low dose rtPA. *International Journal of Stroke* 2011;**6**(SUPPL. 1):14. [DOI: [10.1111/j.1747-4930.2011.00642.x](https://doi.org/10.1111/j.1747-4930.2011.00642.x)]
95. Fuentes S, Huang Y, Wang J, Sharma V, Nguyen HT, Pandian J, et al. The enhanced control of hypertension and thrombolysis stroke Study (ENCHANTED): Part B-rationale for a trial of early intensive blood pressure lowering after use of rtPA. *International Journal of Stroke* 2011;**6**(SUPPL. 1):35. [DOI: [10.1111/j.1747-4930.2011.00643.x](https://doi.org/10.1111/j.1747-4930.2011.00643.x)]
96. Fuentes S, Sharma V, Arima H, Chalmers J, Anderson C. Recanalisation rates of low vs standard dose rtPA in the enhanced control of hypertension and thrombolysis stroke study (ENCHANTED): rationale for a TCD substudy. *Cerebrovascular Diseases (Basel, Switzerland)* 2013;**36**(SUPPL. 1):21. [DOI: [10.1159/000353795](https://doi.org/10.1159/000353795)]
97. Huang Y, Sharma VK, Robinson T, Lindley RI, Chen X, Kim JS, et al. Rationale, design, and progress of the enhanced control of hypertension and thrombolysis stroke study (ENCHANTED) trial: an international multicenter 2x2 quasi-factorial randomized controlled trial of low- vs. standard-dose rt-PA and early intensive vs. guideline-recommended blood pressure lowering in patients with acute ischaemic stroke eligible for thrombolysis treatment. *International Journal of Stroke* 2015;**10**(5):778-88. [DOI: [10.1111/ij.s.12486](https://doi.org/10.1111/ij.s.12486)]
98. Lin Y, Yu J, Tang Y, Tang Q, Wang X, Yang J, et al. Influence of white blood cell count on outcome for thrombolysis treated acute ischemic stroke ENCHANTED (enhanced control of hypertension and thrombolysis stroke study) post hoc analysis. *European Stroke Journal* 2019;**4**(Supplement 1):249. [DOI: [10.1177/2396987319845581](https://doi.org/10.1177/2396987319845581)]
99. Minhas JS, Wang X, Arima H, Bath PM, Billot L, Broderick JP, et al. Lipid-lowering pretreatment and outcome following intravenous thrombolysis for acute ischaemic stroke: a post hoc analysis of the enhanced control of hypertension and thrombolysis stroke study trial. *Cerebrovascular Diseases (Basel, Switzerland)* 2018;**45**(5-6):213-20. [DOI: [10.1159/000488911](https://doi.org/10.1159/000488911)]
100. Minhas J, Wang X, Chalmers J, Anderson C, Robinson T. The enhanced control of hypertension and thrombolysis stroke study (ENCHANTED) blood pressure (BP) arm: effects of timing, target and intensity of systolic-BP control on outcome. *European Stroke Journal* 2019;**4**(Supplement 1):35. [DOI: [10.1177/2396987319845560](https://doi.org/10.1177/2396987319845560)]
101. Minhas JS, Wang X, Lindley RI, Delcourt C, Song L, Woodward M, et al. Comparative effects of intensive-blood pressure versus standard-blood pressure-lowering treatment in patients with severe ischemic stroke in the ENCHANTED trial. *Journal of Hypertension* 2021;**39**(2):280-5. [DOI: [10.1097/HJH.0000000000002640](https://doi.org/10.1097/HJH.0000000000002640)]
102. NCT01422616. Enhanced control of hypertension and thrombolysis stroke study (ENCHANTED). <https://clinicaltrials.gov/study/NCT01422616> 2011.
103. Ren X, Chen C, Wang X, Li Q, Zhao Y, You S, et al. Influence of renal function on blood pressure control and outcome in thrombolized patients after acute ischemic stroke: post-hoc analysis of the ENCHANTED trial. *Frontiers in Endocrinology* 2024;**15**:1341902. [DOI: [10.3389/fendo.2024.1341902](https://doi.org/10.3389/fendo.2024.1341902)]
104. Ren X, Chen C, Zhao Y, Anderson C, Song L. Influence of renal function on blood pressure control and outcomes in thrombolysed patients after acute ischemic stroke: a post hoc analysis of ENCHANTED trial. *European Stroke Journal* 2023;**8**(2 Suppl):174. [DOI: [10.1177/23969873231169660](https://doi.org/10.1177/23969873231169660)]
105. Ren X, Song L, Anderson C, Chen C. Influence of heart disease on outcome for thrombolysis-treated acute ischemic stroke: a post hoc analysis of ENCHANTED trial. *European Stroke Journal* 2023;**8**(2 Suppl):470. [DOI: [10.1177/23969873231169660](https://doi.org/10.1177/23969873231169660)]
106. Ren X, Yang Z, Anderson C, Song L. Prognostic significance of glucose level after acute ischaemic stroke: post hoc analysis of the ENCHANTED trial. *International Journal of Stroke* 2023;**18**(3 Suppl):81. [DOI: [10.1177/17474930231192010](https://doi.org/10.1177/17474930231192010)]
107. Si L, Chen X, Ouyang M, Wang X, Chen G, Cao Y-J, et al. Cost-effectiveness of low-dose compared to standard-dose

alteplase for acute ischemic stroke in China: a within-trial economic evaluation of the ENCHANTED study. *Cerebrovascular Diseases (Basel, Switzerland)* 2023;**52**(2):145-52. [DOI: [10.1159/000525869](https://doi.org/10.1159/000525869)]

108. Song L, Wang X, Moullaali T, Sun L, Yang J, Lindley R, et al. Regional differences in early bp management after acute stroke in international randomized controlled trials of 5000 patients. *International Journal of Stroke* 2020;**15**(1 Suppl):60. [DOI: [10.1177/1747493020963387](https://doi.org/10.1177/1747493020963387)]

109. Song L, Wang X, Robinson T, Lindley R, Arima H, Lavados P, et al. Characteristics, management and response to treatment in Chinese vs. non-Chinese participants in the ENCHANTED trial. *European Stroke Journal* 2017;**2**(1 Suppl):9-10. [DOI: [10.1177/2396987317705236](https://doi.org/10.1177/2396987317705236)]

110. Wang X, Chalmers J, Robinson T, Anderson C. The enhanced control of hypertension and thrombolysis stroke study (ENCHANTED): the interaction between alteplase dose and blood pressure lowering treatment. *European Stroke Journal* 2019;**4**(Suppl 1):5-6. [DOI: [10.1177/2396987319845560](https://doi.org/10.1177/2396987319845560)]

111. Wang X, Li Q, Carcel C, Delcourt C, Anderson C, Woodward M. To treat acute ischemic stroke with standard-dose or low-dose alteplase? An individualized decision making tool. *International Journal of Stroke* 2018;**13**(1 Suppl 1):30. [DOI: [10.1177/1747493018778666](https://doi.org/10.1177/1747493018778666)]

112. Wang X, Li Q, Carcel C, Delcourt C, Anderson C, Woodward M. An individualized decision making tool for use of standard versus low dose alteplase in acute ischemic stroke. *European Stroke Journal* 2018;**3**(1 Suppl 1):102. [DOI: [10.1177/2396987318770127](https://doi.org/10.1177/2396987318770127)]

113. Wang X, Minhas JS, Moullaali TJ, Luca Di Tanna G, Lindley RI, Chen X, et al. Associations of early systolic blood pressure control and outcome after thrombolysis-eligible acute ischemic stroke: results from the ENCHANTED study. *Stroke* 2022;**29**(2):779-87. [DOI: [10.1161/STROKEAHA.121.034580](https://doi.org/10.1161/STROKEAHA.121.034580)]

114. Wang X, Robinson T, Arima H, Broderick J, Demchuk A, Donnan G, et al. Effects of age and ethnicity on the benefits and risks of low-versus standard-dose alteplase in acute ischemic stroke patients: the ENCHANTED trial. *Stroke* 2017;**48**(Suppl 1):tmp25. [DOI: [10.1161/str.48.suppl_1.tmp25](https://doi.org/10.1161/str.48.suppl_1.tmp25)]

115. Wang X, Robinson T, Arima H, Lindley R, Woodward M, Carcel C, et al. Low-dose versus standard-dose alteplase by time from onset to treatment following acute ischemic stroke: the ENCHANTED trial. *European Stroke Journal* 2017;**2**(1 Suppl 1):8. [DOI: [10.1177/2396987317705236](https://doi.org/10.1177/2396987317705236)]

116. Wang X, Robinson T, Chalmers J, Anderson C. The enhanced control of hypertension and thrombolysis stroke study (ENCHANTED) blood pressure (BP) arm: regional differences in blood pressure management. *European Stroke Journal* 2019;**4**(Suppl 1):257. [DOI: [10.1177/2396987319845581](https://doi.org/10.1177/2396987319845581)]

117. Wang X, Robinson T, Lee TH, Arima H, Bath P, Broderick J, et al. Low-dose versus standard-dose alteplase by age, ethnicity, and severity of acute ischaemic stroke: the ENCHANTED trial. *European Stroke Journal* 2017;**2**(1 Suppl):8. [DOI: [10.1177/2396987317705236](https://doi.org/10.1177/2396987317705236)]

118. Wang X, Minhas JS, Moullaali TJ, Di Tanna GL, Lindley RI, Chen X, et al. Associations of early systolic blood pressure control and outcome after thrombolysis-eligible acute ischemic stroke: results from the ENCHANTED study. *Stroke* 2022;**53**(3):779-87. [DOI: [10.1161/STROKEAHA.121.034580](https://doi.org/10.1161/STROKEAHA.121.034580)]

119. Wang X, Song L, Yang J, Sun L, Moullaali TJ, Sandset EC, et al. Interaction of blood pressure lowering and alteplase dose in acute ischemic stroke: results of the enhanced control of hypertension and thrombolysis stroke study. *Cerebrovascular Diseases (Basel, Switzerland)* 2019;**48**(3-6):207-16. [DOI: [10.1159/000504745](https://doi.org/10.1159/000504745)]

120. Wu G, Arima H, Wang J, Yang J, Delcourt C, Coutts SB, et al. Accuracy of body weight estimation for thrombolysis treatment in the ENCHANTED study. *Cerebrovascular Diseases (Basel, Switzerland)* 2014;**37**(Supplement 1):78.

121. Xia C, Wang X, Lindley Richard I, Delcourt C, Chen X, Zhou Z, et al. Early decompressive hemicraniectomy in thrombolysed acute ischemic stroke patients from the international ENCHANTED trial. *Scientific Reports* 2021;**11**(1):16495. [DOI: [10.1038/s41598-021-96087-z](https://doi.org/10.1038/s41598-021-96087-z)]

122. Yang J. Comparative effects of thrombolysed ischemic patients with low platelet count. *Stroke* 2019;**50**(Supplement 1):AWP101. [DOI: [10.1161/str.50.suppl_1.WP101](https://doi.org/10.1161/str.50.suppl_1.WP101)]

123. Yang Ji, Wang X, Chen X, Li Q, Wu G, Lavados P, et al. Impact of clinical trial participation on 'door-to-needle' time performance: the ENCHANTED trial. *Cerebrovascular Diseases (Basel, Switzerland)* 2014;**37**(Supplement 1):457. [DOI: [10.1016/j.jstrokecerebrovasdis.2016.11.038](https://doi.org/10.1016/j.jstrokecerebrovasdis.2016.11.038)]

124. You S, Wang Y, Wang X, Maeda T, Ouyang M, Han Q, et al. Twenty-four-hour post-thrombolysis NIHSS score as the strongest prognostic predictor after acute ischemic stroke: ENCHANTED study. *Journal of the American Heart Association* 2024;**13**(18):e036109. [DOI: [10.1161/JAHA.124.036109](https://doi.org/10.1161/JAHA.124.036109)]

125. Yu J, Wang X, Yang J, Robinson T, Delcourt C, Arima H, et al. Comparative effects of low-dose versus standard-dose alteplase in thrombolysed ischemic patients with low platelet count: the ENCHANTED trial. *Cerebrovascular Diseases (Basel, Switzerland)* 2019;**48**(Supplement 1):107-8. [DOI: [10.1159/000505285](https://doi.org/10.1159/000505285)]

126. Zhang G, Chen C, Ren X, Ouyang M, Billot L, Li Q, et al. Effects of intensive blood pressure lowering on brain oedema in thrombolysed patients with acute ischaemic stroke: enchanted secondary analysis results. *European Stroke Journal* 2023;**8**(2 Suppl):100. [DOI: [10.1177/23969873231169660](https://doi.org/10.1177/23969873231169660)]

127. Zhao Y, Ren X, Anderson C, Song L. The influence of multimorbidity on functional recovery in patients with acute ischaemic stroke: secondary analysis of the enchanted randomized clinical trial. *International Journal of Stroke* 2023;**18**(3 Suppl):34. [DOI: [10.1177/17474930231192010](https://doi.org/10.1177/17474930231192010)]

128. Zhao Y, Ren X, Anderson C, Song L. Effects of multimorbidity on outcome in thrombolysed patients with acute ischaemic stroke: post hoc analysis of ENCHANTED (enhanced control of hypertension and thrombolysis stroke study). *European Stroke Journal* 2023;**8**(2 Supplement):121. [DOI: [10.1177/23969873231169660](https://doi.org/10.1177/23969873231169660)]

- 129.** Zhou Z, Delcourt C, Xia C, Yoshimura S, Carcel C, Torii-Yoshimura T, et al. Low-dose vs standard-dose alteplase in acute lacunar ischemic stroke: the ENCHANTED trial. *Neurology* 2021;**96**(11):E1512-26. [DOI: [10.1212/WNL.0000000000011598](https://doi.org/10.1212/WNL.0000000000011598)]
- 130.** Zhou Z, Delcourt C, Xia C, Yoshimura S, Carcel C, Torii-Yoshimura T, et al. Low dose versus standard dose alteplase in acute lacunar ischemic stroke. *Stroke* 2021;**52**(Suppl 1):mp3. [DOI: [10.1161/str.52.suppl-1.MP3](https://doi.org/10.1161/str.52.suppl-1.MP3)]
- 131.** Zhou Z, Xia C, Mair G, Delcourt C, Yoshimura S, Liu X, et al. Thrombolysis outcomes according to arterial characteristics of acute ischemic stroke by alteplase dose and blood pressure target. *International Journal of Stroke* 2021;**16**(2 Suppl):78. [DOI: [10.1177/17474930211041949](https://doi.org/10.1177/17474930211041949)]
- 132.** Zhou Z, Xia C, Mair G, Delcourt C, Yoshimura S, Liu X, et al. Thrombolysis outcomes according to arterial characteristics of acute ischemic stroke by alteplase dose and blood pressure target. *International Journal of Stroke* 2022;**17**(5):566-75. [DOI: [10.1177/17474930211025436](https://doi.org/10.1177/17474930211025436)]
- 133.** Robinson T, Carr S, Wang X, Lavados P, Olavarria V, Chalmers J, et al. Low-or standard-dose alteplase therapy following acute ischaemic stroke in patients with impaired renal function: a secondary analysis of the ENCHANTED trial. *International Journal of Stroke* 2016;**11**(Suppl 3):47. [DOI: [10.1177/1747493016670567](https://doi.org/10.1177/1747493016670567)]
- 134.** Xia C, Wang X, Lindley Richard I, Delcourt C, Zhou Z, Chen X, et al. Combined utility of blood glucose and white blood cell in predicting outcome after acute ischemic stroke: the ENCHANTED trial. *Clinical Neurology and Neurosurgery* 2020;**198**(7502039):106254. [DOI: [10.1016/j.clineuro.2020.106254](https://doi.org/10.1016/j.clineuro.2020.106254)]
- 135.** Zhou Z, Xia C, Carcel C, Yoshimura S, Wang X, Delcourt C, et al. Intensive versus guideline-recommended blood pressure reduction in acute lacunar stroke with intravenous thrombolysis therapy: the ENCHANTED trial. *European Journal of Neurology* 2021;**28**(3):783-93. [DOI: [10.1111/ene.14598](https://doi.org/10.1111/ene.14598)]
- 136.** Wang X, Song L, Yang J, Sun L, Moullaali TJ, Sandset EC, et al. Interaction of blood pressure lowering and alteplase dose in acute ischemic stroke: results of the enhanced control of hypertension and thrombolysis stroke study. *Cerebrovascular Diseases (Basel, Switzerland)* 2020;**48**(3-6):207-16. [DOI: [10.1159/000504745](https://doi.org/10.1159/000504745)]
- 137.** Wang X, You S, Zhou Z, Delcourt C, Chen X, Wardlaw J, et al. Baseline brain imaging signs in ischemic patients by presence of atrial fibrillation: the ENCHANTED trial. *European Stroke Journal* 2019;**4**(Suppl 1):428. [DOI: [10.1177/2396987319845581](https://doi.org/10.1177/2396987319845581)]
- 138.** Liu F, Chen C, Ren X, Zhang G, Zhang L, Ong S, et al. Associations of early systolic blood pressure control and brain oedema after thrombolysis-eligible acute ischemic stroke: secondary analysis of the ENCHANTED study. *International Journal of Stroke* 2023;**18**(3 Suppl):93-4. [DOI: [10.1177/17474930231192010](https://doi.org/10.1177/17474930231192010)]
- 139.** Minhas J, Wang X, Chalmers J, Anderson C, Robinson T. The enhanced control of hypertension and thrombolysis stroke study (ENCHANTED) blood pressure (BP) arm: comparative effects of intensive-versus standard-bp lowering treatment in large vessel occlusion. *European Stroke Journal* 2019;**4**(Suppl 1):35. [DOI: [10.1177/2396987319845560](https://doi.org/10.1177/2396987319845560)]
- 140.** Chen C, Ren X, Ouyang M, Zhao Y, Wang X, Li Q, et al. Effects of intensive blood pressure lowering on renal function in thrombolysed patients with acute ischaemic stroke: ENCHANTED secondary analysis results. *European Stroke Journal* 2023;**8**(2 Suppl):103. [DOI: [10.1177/23969873231169660](https://doi.org/10.1177/23969873231169660)]
- 141.** Chen G, Wang X, Moullaali T, Delcourt C, Lindley R, Robinson T, et al. Influence of renal impairment on outcome by history of hypertension for thrombolysis-treated acute ischemic stroke: the ENCHANTED trial. *International Journal of Stroke* 2020;**15**(1 Suppl):175. [DOI: [10.1177/1747493020963387](https://doi.org/10.1177/1747493020963387)]
- 142.** Robinson TG, Bray BD, Paley L, Sprigg N, Wang X, Arima H, et al. Applicability of ENCHANTED trial results to current acute ischemic stroke patients eligible for intravenous thrombolysis in England and Wales: comparison with the Sentinel Stroke National Audit Programme registry. *International Journal of Stroke* 2019;**14**(7):678-85. [DOI: [10.1177/1747493019841246](https://doi.org/10.1177/1747493019841246)]
- 143.** Sun L, Song L, Yang J, Lindley Richard I, Robinson T, Lavados Pablo M, et al. Smoking influences outcome in patients who had thrombolysed ischaemic stroke: the ENCHANTED study. *Stroke and Vascular Neurology* 2021;**6**(3):395-401. [DOI: [10.1136/svn-2020-000493](https://doi.org/10.1136/svn-2020-000493)]
- 144.** Yang P, Song L, Zhang Y, Zhang X, Chen X, Li Y, et al. Intensive blood pressure control after endovascular thrombectomy for acute ischaemic stroke (ENCHANTED2/MT): a multicentre, open-label, blinded-endpoint, randomised controlled trial. *The Lancet* 2022;**400**(10363):1585-96. [DOI: [10.1016/S0140-6736\(22\)01882-7](https://doi.org/10.1016/S0140-6736(22)01882-7)]
- 145.** Ren X, Zhao Y, Anderson C, Song L. Influence of multimorbidity on functional outcome in patients with acute ischaemic stroke after endovascular thrombectomy: post hoc analysis of the enchanted2/mt trial. *International Journal of Stroke* 2023;**18**(3 Supplement):81. [DOI: [10.1177/17474930231192010](https://doi.org/10.1177/17474930231192010)]
- 146.** Song L, Yang P, Zhang Y, Zhang X, Chen X, Li Y, et al. The second randomized controlled enhanced control of hypertension and thrombectomy stroke study (ENCHANTED2): protocol and progress. *International Journal of Stroke* 2023;**18**(3):364-9. [DOI: [10.1177/17474930221120345](https://doi.org/10.1177/17474930221120345)]
- 147.** Sun L, Song L, Yang P, Zhang X, Hao J, Cheng G, et al. ENCHANTED2: A trial of intensive blood pressure management after mechanical thrombectomy for acute ischemic stroke. *International Journal of Stroke* 2020;**15**(1 SUPPL):241. [DOI: [10.1177/1747493020963387](https://doi.org/10.1177/1747493020963387)]
- 148.** Zhao Y, Ren X, Anderson C, Song L. The influence of multimorbidity on polypharmacy in patients with acute ischemic stroke after endovascular thrombectomy: post hoc analysis of the enchanted2/mt trial. *International Journal of Stroke* 2023;**18**(3 Suppl):431. [DOI: [10.1177/17474930231201072](https://doi.org/10.1177/17474930231201072)]

- 149.** NCT04140110. Second enhanced control of hypertension and thrombectomy stroke study (ENCHANTED2). <https://clinicaltrials.gov/study/NCT04140110> 2019.
- 150.** ChiCTR1900027785. Second enhanced control of hypertension and thrombectomy stroke study. <https://www.chictr.org.cn/showproj.html?proj=46176> 2020.
- 151.** Ren X, Yang Z, Anderson C, Song L. Influence of multimorbidity on systolic blood pressure in patients with acute ischaemic stroke after endovascular thrombectomy: post hoc analysis of the enchanted2/mt trial. *International Journal of Stroke* 2023;**18**(3 Suppl):279. [DOI: [10.1177/17474930231192010](https://doi.org/10.1177/17474930231192010)]
- 152.** Song L, Yang P, Zhang Y, Zhang X, Li Y, Wang Y, et al. Second, enhanced control of hypertension and thrombectomy stroke study (ENCHANTED-MT): progress report. *European Stroke Journal* 2022;**7**(1 Suppl):165. [DOI: [10.1177/23969873221087559](https://doi.org/10.1177/23969873221087559)]
- 153.** Zhang X, Cao L, Wu S, Wang D, Wang H, Zhang D, et al. Early intensive blood pressure management after endovascular thrombectomy in ischaemic stroke (IDENTIFY): a multicentre, open-label, blinded-endpoint, randomised controlled trial. *The Lancet Regional Health. Western Pacific* 2025;**59**:101589. [DOI: [10.1016/j.lanwpc.2025.101589](https://doi.org/10.1016/j.lanwpc.2025.101589)]
- 154.** Zhang X, Hu F, Hao Z, Ye C, Wan J, Lu K, et al. Efficacy of early intensive blood pressure management after thrombectomy: protocol for a randomized controlled clinical trial (IDENTIFY). *Neurocritical Care* 2023;**38**(1):196-203. [DOI: [10.1007/s12028-022-01618-9](https://doi.org/10.1007/s12028-022-01618-9)]
- 155.** Zhang X, Hu F, Hao Z, Ye C, Pan R, Wu B. Efficacy of early intensive blood pressure management after thrombectomy: protocol for a randomized controlled clinical trial (IDENTIFY). *International Journal of Stroke* 2023;**18**(3 Supplement):167. [DOI: [10.1177/17474930231192010](https://doi.org/10.1177/17474930231192010)]
- 156.** ChiCTR2200057770. Efficacy and safety of ultra-early intensive hypotension after mechanical thrombolectomy in patients with acute anterior circulation ischemic stroke: a prospective, multicenter, randomized controlled, blind outcome evaluation study. <https://www.chictr.org.cn/showprojEN.html?proj=162575> 2022.
- 157.** Nam HS, Kim YD, Heo J, Lee H, Jung Jae W, Choi JK, et al. Intensive vs conventional blood pressure lowering after endovascular thrombectomy in acute ischemic stroke: the OPTIMAL-BP randomized clinical trial. *JAMA* 2023;**330**(9):832-42. [DOI: [10.1001/jama.2023.14590](https://doi.org/10.1001/jama.2023.14590)]
- 158.** Yun J, Kim KH, Jung JW, Kim YD, Heo J, Lee H, et al. Effects of blood pressure lowering in patients treated with intravenous thrombolysis before endovascular thrombectomy. *International Journal of Stroke* 2025;**20**(6):696-707. [DOI: [10.1177/17474930251315630](https://doi.org/10.1177/17474930251315630)]
- 159.** NCT04205305. Outcome in patients treated with intraarterial thrombectomy - optimal blood pressure control (OPTIMAL-BP). <https://clinicaltrials.gov/study/NCT04205305> 2019. [HTTPS://CLINICALTRIALS.GOV/STUDY/: NCT04205305]
- 160.** Kim KH, Yun J, Jung JW, Kim YD, Heo J, Lee H, et al. Intensive blood pressure lowering in acute stroke with intracranial stenosis post-thrombectomy: a secondary analysis of the OPTIMAL-BP trial. *International Journal of Stroke* 2025;**20**(3):328-37. [DOI: [10.1177/17474930241305315](https://doi.org/10.1177/17474930241305315)]
- 161.** Nam HS, Kim YD, Choi JK, Baik M, Kim BM, Kim DJ, et al. Outcome in patients treated with intra-arterial thrombectomy: the optimal blood pressure control (OPTIMAL-BP) trial. *International Journal of Stroke* 2022;**17**(8):931-7. [DOI: [10.1177/17474930211041213](https://doi.org/10.1177/17474930211041213)]
- 162.** Nam HS, Kim YD, Heo J, Lee H, Jung J, Choi JK, et al. Effect of intensive vs conventional blood pressure lowering after successful intra-arterial thrombectomy in acute ischemic stroke: the OPTIMAL-BP randomized clinical trial. *European Stroke Journal* 2023;**8**(2 Suppl):679. [DOI: [10.1177/23969873231174267](https://doi.org/10.1177/23969873231174267)]
- 163.** Jung JW, Kim KH, Yun J, Kim YD, Heo J, Lee H, et al. Functional outcomes associated with blood pressure decrease after endovascular thrombectomy. *JAMA Network Open* 2024;**7**(4):e246878. [DOI: [10.1001/jamanetworkopen.2024.6878](https://doi.org/10.1001/jamanetworkopen.2024.6878)]
- 164.** Jung JW, Kim KH, Yun J, Kim Young D, Heo J, Lee H, et al. Impact of intensive blood pressure lowering after multiple-attempt endovascular thrombectomy: a secondary analysis of the OPTIMAL-BP trial. *International Journal of Stroke* 2024;**19**(10):1134-46. [DOI: [10.1177/17474930241265652](https://doi.org/10.1177/17474930241265652)]
- 165.** Mistry EA, Hart KW, Davis LT, Gao Y, Prestigiacomo CJ, Mittal S, et al. Blood pressure management after endovascular therapy for acute ischemic stroke: the BEST-II randomized clinical trial. *JAMA* 2023;**330**(9):821-31. [DOI: [10.1001/jama.2023.14330](https://doi.org/10.1001/jama.2023.14330)]
- 166.** NCT04116112. Blood pressure after endovascular stroke therapy-II. <http://clinicaltrials.gov/study/NCT04116112> 2019.
- 167.** NCT06352619. Third enhanced control of hypertension and thrombectomy stroke domain within ACT-GLOBAL adaptive platform trial. <https://clinicaltrials.gov/study/NCT06352619> 2024. [HTTPS://CLINICALTRIALS.GOV/STUDY/: NCT06352619]
- 168.** Camps-Renom P, Guasch-Jimenez M, Martinez-Domeno A, Prats-Sanchez L, Ramos-Pachon A, Alvarez-Cienfuegos J, et al. A randomized trial on hemodynamic optimization of cerebral perfusion after successful endovascular therapy in patients with acute ischemic stroke (HOPE). *Cerebrovascular Diseases (Basel, Switzerland)* 2024;**c2g**(9100851):1-8. [DOI: [10.1159/000540606](https://doi.org/10.1159/000540606)]
- 169.** Marti-Fabregas J, Prats-Sanchez L, Martinez-Domeno A, Ramos-Pachon A, Marin R, Guasch-Jimenez M, et al. A randomized trial on hemodynamic optimization of cerebral perfusion after endovascular therapy in patients with acute ischemic stroke (HOPE STUDY). *European Stroke Journal* 2022;**7**(1 SUPPL):152. [DOI: [10.1177/23969873221087559](https://doi.org/10.1177/23969873221087559)]
- 170.** Marti-Fabregas J, Prats-Sanchez L, Jimenez MG, Domeno AM, Pachon AR, Marin R, et al. A randomized trial on hemodynamic optimization of cerebral perfusion after endovascular therapy in patients with acute ischemic stroke (HOPE STUDY). *European Stroke Journal* 2023;**8**(2 Supplement):195EP-196. [DOI: [10.1177/23969873231169660](https://doi.org/10.1177/23969873231169660)]

- 171.** NCT04892511. Hemodynamic optimization of cerebral perfusion after endovascular therapy in patients with acute ischemic stroke. <https://clinicaltrials.gov/study/NCT04892511> 2021. [HTTPS://CLINICALTRIALS.GOV/STUDY/: NCT04892511]
- 172.** NCT05282290. Prevention and treatment of reperfusion injury after mechanical thrombectomy in acute ischemic stroke. <https://clinicaltrials.gov/study/NCT05282290> 2022. [HTTPS://CLINICALTRIALS.GOV/STUDY/: NCT05282290]
- 173.** NCT04775147. Intensive control of blood pressure in acute ischemic stroke after endovascular therapy on clinical outcome (CRISIS I). <https://clinicaltrials.gov/study/NCT04775147> 2021. [HTTPS://CLINICALTRIALS.GOV/STUDY/: NCT04775147]
- 174.** Sharma VK, Tan BY, Sim MY, Kulkarni A, Seow PA, Hong CS, et al. Rationale and design of a randomized trial of early intensive blood pressure lowering on cerebral perfusion parameters in thrombolysed acute ischemic stroke patients. *Medicine* 2018;**97**(40):e12721. [DOI: [10.1097/MD.00000000000012721](https://doi.org/10.1097/MD.00000000000012721)]
- 175.** Sharma V, Tan B, Wong L, Yeo L, Chan B, Teoh HL, et al. Randomized clinical trial of safety and feasibility of early intensive blood pressure lowering in thrombolysed acute ischemic stroke patients and assessed using ct perfusion imaging. *European Stroke Journal* 2019;**4**(Supplement 1):255. [DOI: [10.1177/2396987319845581](https://doi.org/10.1177/2396987319845581)]
- 176.** NCT03443596. Blood pressure control in acute ischemic stroke. <https://clinicaltrials.gov/study/NCT03443596> 2018. [HTTPS://CLINICALTRIALS.GOV/STUDY/: NCT03443596]
- 177.** Zhang Y, Zhou J, Zhang Y, Chen N, Zhou F, Yang J, Wang M, Hou J. Effects of intensive blood pressure lowering on the early reperfusion and prognosis after intravenous thrombolysis in patients with acute ischemic stroke [in Chinese]. *Zhonghua Shenjingke Zazhi (Chinese Journal of Neurology)* 2017;**50**(5):348-353. [DOI: [doi:10.3760/cma.j.issn.1006-7876.2017.05.008](https://doi.org/10.3760/cma.j.issn.1006-7876.2017.05.008)]
- 178.** Bai X, Zhang X, Gong H, Wang T, Wang X, Wang W, et al. Different types of percutaneous endovascular interventions for acute ischemic stroke. *Cochrane Database of Systematic Reviews* 2023, Issue 5. Art. No: CD014676. [DOI: [10.1002/14651858.CD014676.pub2](https://doi.org/10.1002/14651858.CD014676.pub2)]
- 179.** Gottlieb M, Carlson JN, Westrick J, Peksa GD. Endovascular thrombectomy with versus without intravenous thrombolysis for acute ischaemic stroke. *Cochrane Database of Systematic Reviews* 2025, Issue 4. Art. No: CD015721. [DOI: [10.1002/14651858.CD015721.pub2](https://doi.org/10.1002/14651858.CD015721.pub2)]
- 180.** Wardlaw JM, Koumellis P, Liu M. Thrombolysis (different doses, routes of administration and agents) for acute ischaemic stroke. *Cochrane Database of Systematic Reviews* 2013, Issue 5. Art. No: CD000514. [DOI: [10.1002/14651858.CD000514.pub3](https://doi.org/10.1002/14651858.CD000514.pub3)]
- 181.** Jeong HS, Park HK, Park JM, Ko SB, Seo WK, Heo SH, et al. Update of the Korean clinical practice guidelines for stroke: Blood pressure management. *Journal of Clinical Neurology* 2025;**21**(5):370-83. [DOI: [10.3988/jcn.2025.0177](https://doi.org/10.3988/jcn.2025.0177)]
- 182.** Da Silva Neto EP, Ferreira LF, de Cardozo Hernandez ALC, Figueiredo EG, Mota Telles JP. Blood pressure targets after successful reperfusion in mechanical thrombectomy for acute ischemic stroke: an updated systematic review and meta-analysis of randomized clinical trials. *Neurological Sciences* 2024;**45**(8):3879-86. [DOI: [10.1007/s10072-024-07458-z](https://doi.org/10.1007/s10072-024-07458-z)]
- 183.** Zhang K, Wang X, Wu Y, Liang F, Hou X, Zhang Z, et al. Intensive versus standard blood pressure management after endovascular therapy for acute ischemic stroke: a systematic review and meta-analysis. *Journal of Neurosurgical Anesthesiology* 2025;**37**(1):20-30. [DOI: [10.1097/ANA.0000000000000961](https://doi.org/10.1097/ANA.0000000000000961)]
- 184.** Peng TJ, Ortega-Gutiérrez S, de Havenon A, Petersen NH. Blood pressure management after endovascular thrombectomy. *Frontiers in Neurology* 2021;**12**:723461. [DOI: [10.3389/fneur.2021.723461](https://doi.org/10.3389/fneur.2021.723461)]
- 185.** Liao Z, Tian L, Wen M, Wang B, Ding K, Song Q. Safety and efficacy of intensive blood pressure-lowering after successful endovascular therapy in AIS: a meta-analysis. *The Neurologist* 2025;**30**(1):60-7. [DOI: [10.1097/NRL.0000000000000569](https://doi.org/10.1097/NRL.0000000000000569)]
- 186.** Maier B, Delvoye F, Labreuche J, Escalard S, Desilles JP, Redjem H, et al. Impact of blood pressure after successful endovascular therapy for anterior acute ischemic stroke: a systematic review. *Frontiers in Neurology* 2020;**11**:573382. [DOI: [10.3389/fneur.2020.573382](https://doi.org/10.3389/fneur.2020.573382)]
- 187.** Dong X, Liu Y, Chu X, Yu E, Jia X, Ji X, Wu C. Blood pressure management after endovascular thrombectomy: insights of recent randomized controlled trials. *CNS Neuroscience & Therapeutics* 2024;**30**(8):e14907. [DOI: [10.1111/cns.14907](https://doi.org/10.1111/cns.14907)]

ADDITIONAL TABLES

Table 1. Overview of included studies and syntheses: general characteristics

Study	Study design/study date	Setting/country/country classification	Intervention	Comparator	No. of participants randomized (ratio)	Age ^a	Sex ^b	Reperfusion treatment conducted	Outcomes
Avidzba 2024	Parallel group randomized trial March 2022 to August 2024	Setting: inpatient, single-center, national Country: Russian Federation Country classification: high-income country	Intensive systolic blood pressure management target < 160 mmHg	Conventional systolic blood pressure management (< 180 mmHg)	90 (1:1)	Intervention group: 70.2 ± 12.9 Comparator group: 69.4 ± 10.9	Intervention group: 17 (52.5%) Comparator group: 16 (44.4%)	Systemic thrombolysis	Clinical function, clinical function (dichotomous), neurologic adverse events, neurologic adverse events (symptomatic), neurologic status, all-cause mortality, hospital length of stay, other adverse events
BP-TARGET	Parallel group randomized trial June 2017 to January 2020	Setting: inpatient, multicenter, national Country: France Country classification: high-income country	Intensive systolic blood pressure management target < 140 mmHg	Conventional systolic blood pressure management (< 180 mmHg)	324 (1:1)	Intervention group: 77 (66 to 85) Comparator group: 76 (62 to 82)	Intervention group: 77 (49%) Comparator group: 88 (55%)	Endovascular thrombectomy	Clinical function (dichotomous), neurologic adverse events, neurologic adverse events (symptomatic), all-cause mortality, other adverse events
CLEVER	Parallel group randomized trial October 2021 to December 2023	Setting: inpatient, single-center, national Country: United States. Country classification: high-income country	Intensive systolic blood pressure management target < 120 mmHg	Conventional systolic blood pressure management (< 160 mmHg)	80 (1:1)	Intervention group: 74.5 (62.75 to 81.5) Comparator group: 71.5 (64.0 to 82.25)	Over half of the population was female in both groups	Endovascular thrombectomy	Clinical function (dichotomous), neurologic adverse events, neurologic adverse events (symptomatic), all-cause mortality, hospital length of stay, and other adverse events
DETECT	Parallel group randomized trial	Setting: inpatient, single-center, national Country: Canada	Intensive systolic blood pressure management target < 140 mmHg	Conventional systolic blood pressure management	30 (1:1)	Intervention group: 85 (80 to 93) Comparator group: 78 (69 to 82)	Intervention group: 9 (60%)	Endovascular thrombectomy	Clinical function, clinical function (dichotomous), neurologic adverse events, neurologic adverse events (symptomatic), neurologic status, neurologic status dichotomous, all-cause

Table 1. Overview of included studies and syntheses: general characteristics (Continued)

	October 2020 to May 2023	Country classification: high-income country		(< 180 mmHg)		Comparator group: 8 (53%)		mortality, hospital length of stay, other adverse events
ENCHANT-ED	Parallel group randomized trial March 2012 to April 2018	Setting: inpatient, multicenter, multinational Countries: Australia, Brazil, Chile, China, Colombia, Hong Kong, India, Italy, South Korea, Singapore, Spain, Taiwan, Thailand, United Kingdom, Vietnam Country classification: 2 lower-middle-income countries, 5 upper-middle-income countries, and 8 high-income countries	Intensive systolic blood pressure management target < 140 mmHg	Conventional systolic blood pressure management (< 180 mmHg)	2227 (1:1)	Intervention group: 66.7 ± 12.4 Comparator group: 67.1 ± 12.0	Intervention group: 401 (37.1%) Comparator group: 434 (38.9%)	Systemic thrombolysis Clinical function (dichotomous), quality of life, neurologic adverse events (symptomatic), all-cause mortality, hospital length of stay, other adverse events
ENCHANT-ED2/MT	Parallel group randomized trial July 2020 to October 2022	Setting: inpatient, multicenter, national Country: China Country classification: upper-middle-income country	Intensive systolic blood pressure management target < 120 mmHg	Conventional systolic blood pressure management (< 180 mmHg)	821 (1:1)	Intervention group: 68 ± 12.0 Comparator group: 67 ± 12.0	Intervention group: 158 (39%) Comparator group: 152 (37%)	Endovascular thrombectomy Clinical function (dichotomous), quality of life, neurologic adverse events, neurologic adverse events (symptomatic), neurologic status, all-cause mortality, hospital length of stay, other adverse events
IDENTIFY	Parallel group randomized trial October 2022 to March 2024	Setting: inpatient, multicenter, national Country: China Country classification: upper-middle-income country	Intensive systolic blood pressure management target < 140 mmHg	Conventional systolic blood pressure management (< 180 mmHg)	383 (1:1)	Intervention group: 72 ± 11 Comparator group: 72 ± 10	Intervention group: 92 (50.6%) Comparator group: 93 (46.5%)	Endovascular thrombectomy Clinical function, clinical function (dichotomous), neurologic adverse events, neurologic status, neurologic status dichotomous, all-cause mortality, other adverse events

Table 1. Overview of included studies and syntheses: general characteristics (Continued)

OPTI-MAL-BP	Parallel group randomized trial June 2020, to July 2023	Setting: inpatient, multicenter, national Country: Korea Country classification: high-income country	Intensive systolic blood pressure management target < 140 mmHg	Conventional systolic blood pressure management (< 180 mmHg)	306 (1:1)	Intervention group: 73.2 ± 12.1 Comparator group: 72.9 ± 10.8	Intervention group: 63 (40.6%) Comparator group: 60 (40.0%)	Endovascular thrombectomy	Clinical function, clinical function (dichotomous), quality of life, neurologic adverse events, neurologic adverse events (symptomatic), neurologic status, neurologic status dichotomous, all-cause mortality
The BEST-II	Parallel group randomized trial January 2020 to December 2022	Setting: inpatient, multicenter, national Country: United States Country classification: high-income country	Intensive systolic blood pressure management target < 140 mmHg Intensive systolic blood pressure management target < 160 mmHg	Conventional systolic blood pressure management (< 180 mmHg)	120 (1:1:1)	Intervention group 1: 74 (66.8 to 84.8) Intervention group 2: 70.5 (61.8 to 76) Comparator group: 68 (58 to 76.3)	Intervention group 1: 28 (70%) Intervention group 2: 21 (52.5%) Comparator group: 20 (50%)	Endovascular thrombectomy	Clinical function (dichotomous), neurologic adverse events, neurologic adverse events (symptomatic), neurologic status, all-cause mortality, other adverse events

^aMean ± standard deviation (SD) or median (interquartile range (IQR)).

^bNumber of females (percentage of female population).

Table 2. Overview of included studies and syntheses: neurological characteristics

Study	NIHSS ^a	Location of the arterial occlusion ^b	ASPECTS ^c	Treatment performed to achieve reperfusion in stroke ^b	Agent and dose of pharmacologic thrombolytic therapy used	Reperfusion grade after EVT ^b	Time to actual delivery of therapy/re-canalization ^d	Time interval from stroke onset to randomization ^d	Baseline SBP ^d
Avidzba 2024	Total: 8.5 (6; 16.3)	Total: 56 (81.2%) anterior circulation	NR	Total: 69 (100%) systemic thrombolysis 13 (18.8%) EVT	Alteplase 0.9 mg/kg (0.6 mg/kg in pa-	NA	Total: 182.6 ± 44.8 min	NR	< 185 mmHg

Table 2. Overview of included studies and syntheses: neurological characteristics (Continued)

	Intervention group: 12 (6.8; 16.3)			Intervention group: 33 (100%) systemic thrombolysis	tients over 80 years of age)		Intervention group: 186.9 ± 46.9 min		
	Control group: 7 (6; 15.8)			9 (27.3%) EVT Control group: 36 (100%) systemic thrombolysis 4 (11.1%) EVT			Control group: 177.1 ± 42.4 min		
BP-TARGET	Intervention group: 18 (12; 20) Control group: 17 (13; 20)	NR	NR	Intervention group: 85 (54%) systemic thrombolysis 158 (100%) EVT Control group: 88 (52%) systemic thrombolysis 160 (100%) EVT	Alteplase	Intervention group: mTICI 2b: 70 (44%) mTICI 3: 88 (56%) Control group: mTICI 2b: 76 (48%) mTICI 3: 84 (52%)	Intensive SBP target group: 285 min (234; 357) Control group: 297 min (220; 353)	NR	Intervention group: mean 155 mmHg ± 26 Control group: mean 152 mmHg ± 25
CLEVER	Intervention group: 16.5 (11.75; 21.25) Control group: 15.5 (11.75; 21.0)	Intervention group: M1 MCA: 52.5% M2 MCA: 20% Control group: M1 MCA: 32.5% M2 MCA: 42.5%	Intervention group: 10 (9; 10) Control group: 10 (9; 10)	Intervention group: 6 (40%) systemic thrombolysis 40 (100%) EVT Control group: 4 (27%) systemic thrombolysis 40 (100%) EVT	NR	Intervention group: TICI 2c: 16 (40%) TICI 3: 24 (60%) Control group: TICI 2c: 23 (57.5%) TICI 3: 17 (42.5%)	Intensive SBP target group: 273.5 min (209.3; 416.3) Control group: 339 min (225.5; 664.3)	NR	Intervention group: mean 157.4 mmHg ± 26.6 Control group: mean 157.9 mmHg ± 23
DETECT	Intervention group: 19 (15; 22)	Intervention group: ICA: 3 (20%) M1 MCA: 9 (60%)	Intervention group: 8 (7; 10)	Intervention group: 20 (50%) systemic thrombolysis 15 (100%) EVT Control group:	Alteplase	Intervention group: TICI 2b: 3 (20%) TICI 2c: 4 (27%) TICI 3: 8 (53%) Control group:	Intensive SBP target group: 331 min (249; 835)	< 60 minutes	Intervention group: mean 159 mmHg ± 8

Table 2. Overview of included studies and syntheses: neurological characteristics (Continued)

	Control group: 18 (13; 23)	M2 MCA: 3 (20%) Tandem occlusion: 0 (0%) Control group: ICA: 3 (20%) M1 MCA: 8 (53%) M2 MCA: 4 (27%) Tandem occlusion: 2 (13%)	Control group: 7 (6; 8)	18 (45%) systemic thrombolysis 12 (100%) EVT		TICI 2b: 2 (13%) TICI 2c: 6 (40%) TICI 3: 7 (47%)	Control group: 310 min (247; 628)	Control group: Mean 158 mmHg ± 9	
ENCHANT-ED	Intervention group: 7 (4; 12) Control group: 8 (4; 12)	Intervention group: Proximal occlusion: 97 (9%) Control group: Proximal occlusion: 91 (8.2%)	NR	Total: 2227 (100%) systemic thrombolysis 42 (1.9%) EVT Intervention group: 1111 (100%) systemic thrombolysis Control group: 1116 (100%) systemic thrombolysis	Alteplase 0.6 mg /dL or 0.9 mg /dL	NA	NR	Intervention group: 3.4 hours (2.5; 4.1) Control group: 3.3 hours (2.6; 4.1)	Intervention group: mean 165.4 mmHg ± 9.1. Control group: mean 165.2 mmHg ± 9.2
ENCHANT-ED2/MT	Intervention group: 15 (10; 20) Control group: 15 (10; 20)	Intervention group: Anterior circulation occlusion: 264 (81%) Control group: Anterior circulation occlusion: 269 (85%)	NR	Intervention group: 132 (32%) systemic thrombolysis 407 (100%) EVT Control group: 115 (28%) systemic thrombolysis 409 (100%) EVT	Alteplase	Intervention group: TICI 2b: 37 (9%) TICI 2c: 28 (7%) TICI 3: 342 (84%) Control group: TICI 2b: 43 (11%), TICI 2c: 26 (6%), TICI 3: 340 (83%)	NR	Intervention group: 9.3 hours (6.4; 13.6) Control group: 8.6 hours (6.2; 12.2)	Intervention group: mean 159 mmHg ± 15 Control group: mean 160 mmHg ± 14
IDENTIFY	Intervention group: 16 (12; 19)	Intervention group: ICA: 71 (39.0%)	Intervention group: 9 (8; 10)	Intervention group: 77 (42.3%) systemic thrombolysis	NR	Intervention group:	Intervention group:	Intervention group:	NR

Table 2. Overview of included studies and syntheses: neurological characteristics (Continued)

	Control group: 15 (12; 15)	M1 MCA: 80 (44.0%) M2 MCA: 27(14.8%) Other sites: 4(2.2%) Control group: ICA: 66 (33.2%) M1 MCA: 111 (55.8%) M2 MCA: 22 (11.1%) Other sites: 0 (0.0%)	Control group: 9 (8; 10)	183 (100%) EVT Control group: 85 (42.5%) systemic thrombolysis 200 (100%) EVT		TICI 0 to 2a: 3 (1.7%) TICI 2b: 35 (19.2%) TICI 3: 144 (79.1%) Control group: TICI 0 to 2a: 0 (0%) TICI 2b: 35 (17.8%) TICI 3: 162 (82.2%)	4.88 hours (3.95; 6.08) Control group: 4.55 hours (3.48; 5.73)	6.19 hours (5.15; 7.40) Control group: 5.98 hours (4.69; 7.13)	
OPTI-MAL-BP	Intervention group: NIHSS 0 to 5: 14 (9.0%); 6 to 15: 83 (53.5%); ≥ 16: 58 (37.4%) Control group: NIHSS 0 to 5: 18 (12.0%); 6 to 15: 81 (54.0%); ≥ 16: 51 (34.0%)	Intervention group: ICA: 34 (21.9%) MCA: 108 (69.7%) ACA: 1 (0.6%) BA: 12 (7.7%) VA: 4 (2.6%) PCA: 0 (0.0%) Control group: ICA: 32 (21.3%) MCA: 101 (67.3%) ACA: 3 (2.0%) BA: 12 (8.0%) VA: 3 (2.0) PCA: 1 (0.7%)	Intervention group: 9 (7; 10) Control group: 9 (8; 10)	Intervention group: 44 (28.4%) systemic thrombolysis 155 (100%) EVT Control group: 54 (36.7%) systemic thrombolysis 150 (100%) EVT	Alteplase	Intervention group: TICI 2b: 34 (21.9%) TICI 2c: 18 (11.6%) TICI 3: 103 (66.5%) Control group: TICI 2b: 34 (22.7%) TICI 2c: 22 (14.7%) TICI 3: 94 (62.7%)	Intervention group: 421.0 min (265.5; 772.0) Control group: 395.5 min (253.0; 764.0)	Intervention group: 495.0 min (327.5; 825.0) Control group: 472.5 min (315.0; 820.0)	Intervention group: mean 155.2 mmHg ± 13.4 Control group: mean 154.8 mmHg ± 14.4
The BEST-II	Intervention group 1: 16 (13; 23)	Intervention group 1: ICA 7 (17.5%)	Intervention group 1: 8 (7; 9)	Intervention group 1: 17 (42.5%) systemic thrombolysis	NR	Intervention group 1: TICI 2b: 16 (40%)	NR	NR	Intervention group: mean 150

Table 2. Overview of included studies and syntheses: neurological characteristics (Continued)

Intervention group 2: 18 (15; 23)	M1 MCA 31 (77.5%)	Intervention group 2: 7 (6; 8)	40 (100%) EVT	TICI 2c: 6 (15%)	mmHg ± 22.5
Control group: 14 (11; 17)	M2 MCA 6 (15%)	Control group: 8 (7; 9)	Intervention group 2: 19 (47.5%) systemic thrombolysis	TICI 3: 18 (45%)	Intervention group: mean 151 mmHg ± 25.3
	Intervention group 2: ICA 9 (22.5%)		Control group: 18 (45%) systemic thrombolysis	TICI 2b: 17 (42.5%)	Control group: mean 146 mmHg ± 23
	M1 MCA 25 (62.5%)		40 (100%) EVT	TICI 2c: 5 (12.5%)	
	M2 MCA 8 (20%)			TICI 3: 18 (45%)	
	Control group: ICA: 5 (12.5%)			Control group: TICI 2b: 17 (42.5%)	
	M1 MCA 23 (57.5%)			TICI 2c: 7 (17.5%),	
	M2 MCA 15 (37.5%)			TICI 3: 16 (40%)	

a. Median (IQR) or N (%); b. Number of participants (%); c. Median (IQR); d. Median (IQR) or mean ± SD

ACA: anterior cerebral artery; ASPECTS: Alberta Stroke Program Early CT Score; BA: basilar artery; EVT: endovascular thrombectomy; ICA: internal carotid artery; IQR: interquartile range; kg: kilograms; M1 MCA: first segment of the middle cerebral artery; M2 MCA: second segment of the middle cerebral artery; MCA: middle cerebral artery; mg: milligrams; min: minutes; mTICI: the modified Thrombolysis In Cerebral Infarction; NA: not applicable; NIHSS: National Institutes of Health Stroke Scale; NR: not reported; PCA: posterior cerebral artery; SBP: systolic blood pressure; SD: standard deviation; TICI: Thrombolysis In Cerebral Infarction; VA: vertebral artery